Indexed Bellman Information Complexity

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Abstract

We develop *indexed Bellman information complexity*, a representation-level theory of interactive decision making centered on information indices and reference histories. The representation strips away problem-specific syntax and retains only the ingredients needed for dynamic programming and information accounting, thereby unifying the earlier framework of indexed algorithmic information ratios (AIR). On the upper-bound side, regret is controlled by Bellman supersolutions or potential identities whose gradient bracket is paid for by indexed information. Upper-confidence-bound (UCB), estimation-to-decision/decision-estimation-coefficient (E2D/DEC), and adaptive-minimax-sampling or exploration-by-optimization (AMS/EBO) methods appear as three relaxations of this same identity. On the lower-bound side, the posterior-reference trajectory supplies both the information telescope and the ghost quantile of small-regret trajectories. The resulting critical radius in the lower bound is an effective-dimension-scale quantity, as in Fano and local-prior-mass lower bounds, rather than the constant radius of a two-point Le Cam argument. The examples show that DEC is best viewed as a one-step relaxation of indexed Bellman information complexity, not as a universally tight conversion mechanism. We illustrate the framework through several applications, with particular emphasis on kernel bandits. In this setting, the finite active-action marginal provides a concrete basis for comparing UCB, E2D, and AMS/EBO.

Contents

1 Introduction

Information-theoretic lower bounds compare two quantities. The first is the information revealed by an experiment. The second is the mass, entropy, or packing size of the set on which a learner can be successful. In noninteractive statistical estimation this comparison is especially transparent: the experiment is fixed in advance, and local-prior-mass or metric-entropy calculations can be balanced against Kullback-Leibler (KL) information; see, e.g., the information-theoretic upper and lower bounds of Zhang (2006); Yu (1997). In interactive decision making the learner designs the experiment while learning. The action distribution is itself a random consequence of the unknown environment through the history. This adaptivity is exactly what makes a sharp theory difficult.

A powerful response is algorithmic Fano's method, meaning a Fano-type inequality in which the comparison data set is generated from an algorithm-dependent reference or "ghost" experiment rather than from a fixed nonadaptive sampling scheme. This perspective appears in recent interactive Fano frameworks (Chen et al., 2024) and algorithmic lower bounds (Xu, 2026); it is also closely related to the decision-estimation coefficient (DEC) approach of Foster et al. (2021) and its constrained/localized refinements (Foster et al., 2023). The central obstacle in the algorithmic Fano method is what comes next: collapsing a T-round adaptive experiment into a single one-round action distribution generally requires a localized change-of-measure argument, because the action law under a hard environment need not coincide with the action law under the reference experiment. Such a reduction can obscure or discard the geometry of the posterior trajectory, making the theory more rigid than necessary.

This paper leverages indexed posterior-reference histories to construct a lower-bound counterpart of the information telescopes appearing in the upper-bound theory. Given a prior μ, the Bayesian mixture law generates a reference history H'_T and a posterior process. The indexed information gain is evaluated on that same reference history. If $Y = \chi(\Omega)$ is the chosen information index, then the exact identity is

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T} \sum_{t=1}^{T} \mathcal{I}_\chi(S'_t, p'_t),$$

where S'_t is the reference posterior state and p'_t is the algorithm's action distribution at that reference history. The same reference history defines the ghost probability that the algorithm has small target-wise average regret. Thus, the lower-bound theorem takes a stepwise form that mirrors the algorithmic information ratio (AIR) and model-index algorithmic information ratio (MAIR) telescopes in our upper-bound theorems.

$$\text{if} \qquad T\bar{C}^{\mathsf{Alg}}_\chi(\mu) \le \mathrm{kl}\left(\frac{1}{2}, p^{\mathsf{Alg}}_r(\mu,\chi)\right), \qquad \text{then} \qquad \mathbb{E}^{\mathsf{Alg}}_\mu \operatorname{Reg}_T \ge \frac{Tr}{2}.$$

Here r is an average-regret threshold, p^{Alg}_r is the posterior-reference ghost probability of average regret at most r, and $\bar{C}^{\mathsf{Alg}}_\chi = T^{-1}\mathbb{E}\sum_t \mathcal{I}_\chi(S'_t, p'_t)$ is average indexed information. The average notation changes nothing mathematically; it displays the horizon explicitly. The critical balance is

$$T\bar{C}_\chi \asymp \log \frac{1}{p_r}.$$

When the ghost-good mass is roughly $\exp(-d_{\text{eff}})$, the admissible total information is d_{eff}-level. This is the Fano/local-entropy radius, not the constant two-point radius. In other words, we identify the posterior-reference indexed-information coordinates in which interactive Fano and AIR/MAIR upper bounds become two sides of the same Bellman information complexity.

The word "Bellman" in the title is deliberate. The exact value of an interactive problem is a dynamic program, following Bellman's principle of optimality (Bellman, 1957). An upper bound is a Bellman supersolution, or admissible relaxation in the sense of Rakhlin et al. (2012): it lies above the dynamic value and gives an algorithm. A lower bound is a Bellman subsolution or, more flexibly, a Bellman-Fano certificate that combines an information supersolution with a ghost-good-mass supersolution. The one-round DEC is recovered only after replacing the dynamic continuation value by a worst-case offset against a static reference. It may be sharp in simple geometries, but it is not the minimal object for matching upper and lower bounds.

Main contributions.

1. We formulate *indexed Bellman experiments*, a representation-level framework in which the state is sufficient for losses, observation kernels, posterior predictive laws, posterior updates, and indexed information increments. This subsumes DMSO as a special case (Foster et al., 2021) and explores the minimal structural specifications for sequential learnability.

2. We prove the indexed posterior-reference chain rule and the AIR/MAIR regret identities. AIR corresponds to the decision index $Y = \pi^\star(\Omega)$; MAIR corresponds to the model or environment index $Y = \Omega$. We record simplifications and specializations, the gradient brackets, and the reason Danskin's theorem appears in the original AIR proof of Xu and Zeevi (2025).

3. We give a unified "one identity, three algorithms" theorem. UCB methods (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010) certify the bracket through self-normalized calibration and optimism. E2D/DEC methods (Foster et al., 2021, 2023) optimize a one-step offset. AMS/EBO methods (Lattimore and György, 2021; Xu and Zeevi, 2025) control a robust saddle version of the same bracket.

4. We prove the reference-history quantile lower theorem and Bellman lower certificates. As an indexed posterior-reference specialization of the interactive Fano method, the lower side uses the same stepwise indexed information as the upper side, but compares it to ghost-good entropy rather than to a one-round coefficient. Through bandit examples we show DEC/E2D is best viewed as a one-step relaxation of indexed Bellman information complexity, not as a universally tight conversion coefficient.

5. We develop several representative examples, including kernel bandits with an indexable action set, multi-armed bandits, linear bandits, and a Bellman-rank embedding. The kernel-bandit example highlights a regime where the optimal active action is finite, even though the unknown RKHS function remains infinite-dimensional. This makes indexed AIR or optimism the natural theory, while global DEC may be vacuous unless it is properly localized.

Organization. Section 2 defines the dynamic foundation and indexed Bellman experiments. Section 3 gives the indexed information identities and upper algorithmic relaxations. Section 4 gives the quantile lower theorem, Bellman-Fano certificates, and the status of one-round DEC. Section 5 gives applications to kernel bandits and other problems.

2 Indexed Bellman experiments and the dynamic foundation

2.1 Dynamic programming before representation

The minimal representation is easiest to identify from the dynamic program. Fix a prior μ over a latent environment Ω. At time t, after history

$$H_{t-1} = (A_1, O_1, \ldots, A_{t-1}, O_{t-1}),$$

the learner chooses a distribution $p_t(\cdot \mid H_{t-1})$ over available actions and then samples A_t from it. The observation is drawn from a kernel

$$O_t \sim P_{\omega,t}(\cdot \mid H_{t-1}, A_t).$$

The one-step loss is denoted by $\ell_t(\omega, H_{t-1}, A_t)$. Throughout the lower-bound results, we assume that ℓ_t is nonnegative, as is standard in lower-bound arguments for stochastic regret. This assumption is not a modeling restriction for the upper-bound analyses, which also accommodate model misspecification and adversarial losses: one may simply choose a nonnegative surrogate loss. The goal of reinforcement learning and interactive decision making, in their most general form, is to optimize over admissible policies that map histories to actions at each round so as to minimize cumulative loss, in accordance with Bellman's principle of optimality (Bellman, 1957).

The Bayesian posterior is $b_t = \mathcal{L}_\mu(\Omega \mid H_{t-1})$. If a state s is sufficient to evaluate losses, posterior-predictive laws, and posterior updates; see Definition 2.3 below for a rigorous formulation. When ℓ denotes the instantaneous regret, or a nonnegative loss after subtracting the chosen benchmark, the exact Bayes-regret value satisfies

$$V_{T+1}(s) = 0, \qquad V_t(s) = \inf_{p \in \Delta(\mathcal{A}(s))} \left\{ \ell(s,p) + \mathbb{E}_{a\sim p,\, o \sim P_{s,a}} V_{t+1}(\tau(s,a,o)) \right\}. \tag{1}$$

Here $\mathcal{A}(s)$ is the action set at state s, τ is the state update, and

$$\ell(s,p) = \mathbb{E}_{a\sim p} \int \ell_\omega(s,a) b_s(d\omega)$$

is the posterior expected one-step regret/loss. Thus the object is both a dynamic program and a regret-minimization problem: the dynamic program is the exact Bayes regret minimization problem for the chosen prior and representation. Minimax regret is connected through Yao's principle (Yao, 1977):

$$\mathfrak{R}_T^\star := \inf_{\mathsf{Alg}} \sup_{\omega \in \Omega} \mathbb{E}_\omega^{\mathsf{Alg}} L_\omega(H_T) \geq \sup_{\mu} \inf_{\mathsf{Alg}} \mathbb{E}_{\omega\sim\mu} \mathbb{E}_\omega^{\mathsf{Alg}} L_\omega(H_T).$$

Definition 2.1 (Bellman supersolution and subsolution). *Let T_t be the Bellman operator in (1). A sequence $\overline{V}_t$ is a regret supersolution if*

$$\overline{V}_{T+1} \geq 0, \qquad \overline{V}_t(s) \geq (\mathsf{T}_t \overline{V}_{t+1})(s) \quad \textit{for all } t, s.$$

A sequence $\underline{V}_t$ is a regret subsolution if

$$\underline{V}_{T+1} \leq 0, \qquad \underline{V}_t(s) \leq (\mathsf{T}_t \underline{V}_{t+1})(s) \quad \textit{for all } t, s.$$

Proposition 2.2 (Bellman sandwich). *If $\overline{V}$ is a supersolution, then any policy that attains the right-hand side of the supersolution inequality up to errors ε_t has Bayes regret at most $\overline{V}_1(s_1) + \sum_t \varepsilon_t$. If $\underline{V}$ is a subsolution, then every algorithm has Bayes regret at least $\underline{V}_1(s_1)$. Consequently,*

$$\underline{V}_1(s_1) \le V_1(s_1) \le \overline{V}_1(s_1).$$

Proof. For the upper statement, if the selected decision kernel satisfies

$$\ell(s, p_t) + \mathbb{E}\overline{V}_{t+1}(S_{t+1}) \le \overline{V}_t(s) + \varepsilon_t,$$

then summing over t telescopes the potential and yields the claimed bound. The lower statement is the same induction with the inequality reversed: for any decision kernel p_t,

$$\ell(s, p_t) + \mathbb{E}\underline{V}_{t+1}(S_{t+1}) \ge \underline{V}_t(s).$$

Summing gives expected loss at least $\underline{V}_1(s_1) - \mathbb{E}\underline{V}_{T+1}(S_{T+1}) \ge \underline{V}_1(s_1)$. □

This proposition is the organizing principle of the paper. The upper theory constructs supersolutions by paying the Bellman bracket with information. The lower theory constructs subsolutions indirectly, by showing that any algorithm with too little information has too small a ghost-good probability to achieve low regret. The representation required by both sides is exactly the representation in which these Bellman backups are well-defined.

2.2 Latent environments, indices, and losses

The latent environment space is denoted by Ω. An information index is a measurable map

$$Y = \chi(\Omega) \in \mathcal{Y}.$$

The model-index choice $Y = \Omega$ gives MAIR. The action-index choice $Y = A^\star(\Omega)$, where $A^\star$ is an optimal action under the environment, gives AIR. The loss may depend on more than Y. For index-based lower bounds we therefore use the conditional loss

$$\ell_t^\chi(y, H_{t-1}, a) := \mathbb{E}_\mu[\ell_t(\Omega, H_{t-1}, a) \mid \chi(\Omega) = y, H_{t-1}], \tag{2}$$

assuming the relevant regular conditional distributions exist. Define

$$L_\omega(H_T) = \sum_{t=1}^{T} \ell_t(\omega, H_{t-1}, A_t), \qquad \bar{L}_\omega(H_T) = T^{-1} L_\omega(H_T),$$

and

$$L_\chi(y, H_T) = \sum_{t=1}^{T} \ell_t^\chi(y, H_{t-1}, A_t), \qquad \bar{L}_\chi(y, H_T) = T^{-1} L_\chi(y, H_T).$$

Under the true Bayesian experiment,

$$\mathbb{E}_{\Omega\sim\mu, H_T\sim\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) = \mathbb{E}_{Y,H_T} L_\chi(Y, H_T). \tag{3}$$

The choice of information index is fairly flexible. If a pathwise, model-wise event is needed, one may take $Y = \Omega$. If the objective is decision identification, one may take $Y = A^\star(\Omega)$ and pay only for information about the optimal decision. If a value-function class $\mathcal{F}$ provides Bellman sufficiency under the chosen state representation, one may take $Y = f^\star(\Omega)$ and pay only for information about the optimal value function.

2.3 Bellman-sufficient representation

We work with an indexed sequential experiment. The environment $\omega \in \Omega$ is fixed in the frequentist problem. A prior μ over Ω is used only when we discuss Bayes values, posterior-reference trajectories, or Yao/Fano lower bounds. At round t, the learner observes a Bellman state S_t, chooses $A_t \in \mathcal{A}(S_t)$, observes $O_t \sim P_{\omega,t}(\cdot \mid S_t, A_t)$, and incurs a nonnegative per-round regret/loss $\ell_\omega(S_t, A_t) \in [0, L]$ (for upper-bound analyses with misspecification or adversarial losses, one can simply choose a nonnegative surrogate loss). The next state is $S_{t+1} = \tau_t(S_t, A_t, O_t)$.

The full history is always an admissible state. A compressed state is allowed when it is Bellman sufficient: it preserves the loss, the predictive law, the state update, and the indexed information increment. In Bayesian bandits the posterior over models is such a state. In UCB analyses the state may be an estimator and confidence set. In low-Bellman-rank contextual decision processes (CDPs) the state may include a function-class or witness representation of Bellman error.

The framework separates the statistical experiment from the proof device. It is best viewed as a frequentist-minimax framework equipped with optional Bayesian proof devices. The minimax objective is frequentist: the environment ω is fixed. Priors and posteriors are used either as algorithmic beliefs or as reference measures for Bayes/Yao/Fano arguments. A Bayesian upper bound becomes a frequentist upper bound only after a calibration or domination step showing that the algorithmic belief controls the fixed truth. In summary, the framework is a frequentist indexed sequential experiment equipped with optional Bayesian/posterior-reference representations for information accounting. Algorithmic beliefs and posteriors, DMSO estimators, UCB confidence states, and Bellman-rank witnesses are different Bellman-sufficient representations of that experiment.

Definition 2.3 (Indexed Bellman-sufficient representation)**.** *Fix a prior μ and an index map χ. Let $b_t = \mathcal{L}_\mu(\Omega \mid H_{t-1})$ be the posterior under the Bayesian mixture law. A representation is a process*

$$S_t = \phi_t(H_{t-1}, b_t).$$

The time index may be included in S_t; equivalently, the kernels and updates may be time-inhomogeneous, and we suppress the subscript t when it is part of the state. It is exact Bellman-sufficient if, for every state s, action $a \in \mathcal{A}(s)$, and environment ω, there exist objects

$$P_{\omega,s,a} \in \Delta(\mathcal{O}), \qquad \ell_\omega(s,a) \in [0, L], \qquad \tau(s,a,o),$$

such that the following hold.

(i) Predictive sufficiency: $P_{\omega,t}(\cdot \mid H_{t-1}, a) = P_{\omega,S_t,a}$.

(ii) Loss sufficiency: $\ell_t(\omega, H_{t-1}, a) = \ell_\omega(S_t, a)$.

(iii) Posterior predictive sufficiency: $P_{s,a} := \int P_{\omega,s,a}\, b_s(d\omega)$ *where* $b_s := \mathcal{L}_\mu(\Omega \mid S_t = s)$, *is determined by* s.

(iv) Update sufficiency: *under the posterior-reference law,* $S_{t+1} = \tau(S_t, A_t, O_t)$.

(v) Index sufficiency: *for* $q_s = \mathcal{L}(\chi(\Omega) \mid S_t = s)$ *and* $b_s^y = \mathcal{L}(\Omega \mid S_t = s, \chi(\Omega) = y)$ *defined for* q_s*-almost every* y*, the conditional predictive laws*

$$P_{s,a}^y := \int P_{\omega,s,a}\, b_s^y(d\omega)$$

and conditional losses

$$\ell^X_y(s,a) := \int \ell_\omega(s,a) b^y_s(d\omega)$$

are determined by s.

The state should be understood as a controlled sufficient statistic for both fundamental information accounting and practical algorithm design. Classical sufficiency starts from a statistic T, namely a random variable constructed from the data X alone and not depending on the unknown parameter θ. The statistic T is sufficient for the model $\{P_\theta : \theta \in \Theta\}$ if, after conditioning on T, the remaining conditional distribution of the data X is independent of θ (Fisher, 1922). Examples and characterizations include posterior distributions, factorization criteria, and exponential-family descriptions (Xu and Zeevi, 2025). Here we need a decision-theoretic analogue: a statistic is sufficient if it preserves everything needed for future prediction, loss, information, and updating. The state can be the full history, the full posterior, or a compressed feature. The full history is always sufficient but rarely useful. A posterior is sufficient in ordinary Bayesian bandits. A Gaussian mean-covariance pair is sufficient for finite-dimensional Gaussian linear bandits. In finite-action kernel bandits, the active marginal posterior mean and covariance on the active action set are sufficient for the finite experiment, even though the unknown function remains an RKHS element. For low Bellman-rank contextual decision processes, a valid state must include the witness features, value-function class, roll-in distribution, and any posterior or confidence object needed to predict future Bellman errors; bounded Bellman rank alone is not a model class.

DMSO (Foster et al., 2021) is one instantiation: Ω is a model M, an action is a decision π, $P_{M,\pi}$ is the observation law, and the state may be an estimator, confidence set, Gaussian belief, or other calibrated representation. However, the DMSO formalism itself is not required for, nor intrinsic to, tight upper and lower bounds in indexed sequential experiments. The indexed Bellman framework in Definition 2.3 is more primitive. Its true requirement is Bellman sufficiency: whether the chosen state is sufficient for prediction, updating, loss evaluation, and information accounting. Approximate representations can be used by adding explicit approximation errors to the Bellman inequalities.

This is aligned with the representational philosophy behind low Bellman rank for contextual decision processes (Jiang et al., 2017). In particular, contextual decision processes with low Bellman rank provide an important example of representation-level structure. Bellman rank is not a posterior over models; it is a factorization of Bellman errors relative to a chosen function class, roll-in policy family, and sampling scheme. Therefore bounded Bellman rank is not, by itself, a DMSO class or an intrinsic DEC value. It becomes an indexed Bellman experiment only after one fixes the latent environment, the index, the observable trajectory law, the loss, and the Bellman-error witness class whose estimation cost is being charged. Under that additional structure, the Bellman-rank state can be viewed as a compressed Bellman-sufficient representation, analogous to but distinct from a Bayesian posterior state.

3 Indexed information and exact AIR/MAIR identities

3.1 Posterior-reference histories and index information

For a prior μ and algorithm Alg, define the Bayesian posterior-reference trajectory law

$$\bar{\mathbb{P}}^{\mathsf{Alg}}_\mu := \int \mathbb{P}^{\mathsf{Alg}}_\omega \, \mu(d\omega).$$

A reference history is denoted

$$H'_T \sim \bar{\mathbb{P}}_\mu^{\mathsf{Alg}}, \qquad S'_t = \phi_t(H'_{t-1}, b'_t), \qquad b'_t = \mathcal{L}_\mu(\Omega \mid H'_{t-1}),$$

and the algorithm's reference decision kernel is

$$p'_t = p_t(\cdot \mid H'_{t-1}).$$

Definition 3.1 (Indexed information gain)**.** *At state s, let q_s be the posterior law of $Y = \chi(\Omega)$ and let $P^y_{s,a}$ and $P_{s,a}$ be the conditional and marginal predictive laws from Definition 2.3. For $p \in \Delta(\mathcal{A}(s))$, define*

$$\mathcal{I}_\chi(s,p) := \mathbb{E}_{a\sim p} \int D_{\mathrm{KL}}(P^y_{s,a} \| P_{s,a}) q_s(dy). \tag{4}$$

When $Y = \Omega$, this is model-index information. When $Y = \pi^\star(\Omega)$, this is action-index information. In the action-index case, the belief is still a belief over environments; only the information target is the optimal action.

Define the cumulative and average indexed information of an algorithm by

$$C_\chi^{\mathsf{Alg}}(\mu) := \mathbb{E}_{H'_T\sim\bar{\mathbb{P}}_\mu^{\mathsf{Alg}}} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t), \qquad \bar{C}_\chi^{\mathsf{Alg}}(\mu) := \frac{1}{T} C_\chi^{\mathsf{Alg}}(\mu). \tag{5}$$

Proposition 3.2 (Reference-history indexed chain rule)**.** *For every prior μ, index map χ, and algorithm Alg,*

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T\sim\bar{\mathbb{P}}_\mu^{\mathsf{Alg}}} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t) = T\bar{C}_\chi^{\mathsf{Alg}}(\mu). \tag{6}$$

Proof. The mutual-information chain rule gives

$$I_\mu(Y; H_T) = \sum_{t=1}^T I_\mu(Y; A_t, O_t \mid H_{t-1}).$$

Given H_{t-1}, the action A_t is sampled by the algorithm using only randomization independent of the environment and is conditionally independent of Y. Therefore

$$I_\mu(Y; A_t, O_t \mid H_{t-1}) = I_\mu(Y; O_t \mid H_{t-1}, A_t).$$

Conditioning further on $A_t = a$, the law of O_t given $Y = y$ is $P^y_{S_t,a}$, while the posterior predictive law is $P_{S_t,a}$. Hence

$$I_\mu(Y; O_t \mid H_{t-1}, A_t = a) = \int D_{\mathrm{KL}}(P^y_{S_t,a} \| P_{S_t,a}) q_{S_t}(dy).$$

Averaging over $a \sim p_t(\cdot \mid H_{t-1})$ and over the marginal law of the history under the Bayesian mixture law proves the result. Writing this marginal law as the reference history law gives the displayed expression. ∎

3.2 Fixed-truth log gain and deterministic specialization

The information gain in (4) is posterior-averaged. The regret identity used in frequentist upper bounds is slightly finer: it fixes the true environment and tracks the posterior log ratio of its index.

Fix a state s, an environment ω, and let $y = \chi(\omega)$. Assume domination and $q_s(y) > 0$. Define

$$\Lambda_\chi(s, a, o; y) := \log \frac{dP^y_{s,a}}{dP_{s,a}}(o).$$

For $p \in \Delta(\mathcal{A})$, define the fixed-truth index log gain

$$J_\chi(s, p; \omega) := \mathbb{E}_{a\sim p,\, O\sim P_{\omega,s,a}} \Lambda_\chi(s, a, O; \chi(\omega)). \tag{7}$$

Equivalently,

$$J_\chi(s, p; \omega) = \mathbb{E}_{a\sim p} \left[D_{\mathrm{KL}}(P_{\omega,s,a} \| P_{s,a}) - D_{\mathrm{KL}}(P_{\omega,s,a} \| P^{\chi(\omega)}_{s,a}) \right]. \tag{8}$$

Thus the fixed-truth log gain equals the indexed information only after posterior averaging over ω conditional on the index, or in special cases in which the true conditional law already equals the conditional predictive component. In general, the residual

$$D_{\mathrm{KL}}(P_{\omega,s,a} \| P^{\chi(\omega)}_{s,a})$$

is a calibration term. This is why UCB proofs need both self-normalized calibration and optimism: optimism turns regret into a width, and calibration turns the fixed truth into the posterior-reference information geometry.

For $\eta > 0$, define the fixed-truth indexed bracket

$$B_{\chi,\eta}(s, p; \omega) := \ell_\omega(s, p) - \frac{1}{\eta} J_\chi(s, p; \omega), \qquad \ell_\omega(s, p) := \mathbb{E}_{a\sim p} \ell_\omega(s, a). \tag{9}$$

Corollary 3.3 (Deterministic-action indexed regret identity)**.** *Suppose the algorithm is deterministic after conditioning on H_{t-1}, so $p_t = \delta_{a_t}$. Fix $\omega^\star$ and $y^\star = \chi(\omega^\star)$. If the reference index posterior is updated by Bayes' rule, then conditionally on H_{t-1},*

$$\ell_{\omega^\star}(S_t, a_t) = B_{\chi,\eta}(S_t, \delta_{a_t}; \omega^\star) + \frac{1}{\eta} \mathbb{E}_{O_t\sim P_{\omega^\star,S_t,a_t}} \left[\log \frac{q_{t+1}(y^\star)}{q_t(y^\star)} \middle| H_{t-1} \right]. \tag{10}$$

Consequently,

$$\mathbb{E}^{\mathsf{Alg}}_{\omega^\star} L_{\omega^\star}(H_T) = \frac{1}{\eta} \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \log \frac{q_{T+1}(y^\star)}{q_1(y^\star)} + \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} B_{\chi,\eta}(S_t, \delta_{a_t}; \omega^\star). \tag{11}$$

The outer expectation is only over the observation process and any exogenous randomness. In a deterministic noiseless experiment the same identity is pathwise.

Proof. Bayes' rule gives

$$\log \frac{q_{t+1}(y^\star)}{q_t(y^\star)} = \log \frac{dP^{y^\star}_{S_t,a_t}}{dP_{S_t,a_t}}(O_t).$$

Taking conditional expectation under the fixed truth $\omega^\star$ gives $J_\chi(S_t, \delta_{a_t}; \omega^\star)$. Rearranging the definition of $B_{\chi,\eta}$ proves the one-step identity. Summing over t telescopes the posterior log ratio. $\square$

This corollary is the deterministic specialization requested in UCB analyses. It drops the outer expectation over randomized decision laws; what remains is the posterior-ratio telescope and the fixed-truth gradient bracket.

3.3 Exact AIR and MAIR regret identities

We now record the finite form used in the AIR/MAIR literature (Xu and Zeevi, 2025). Although we use the DMSO language to state the AIR/MAIR results and facilitate comparison with earlier work, the results in this subsection require only the Bellman-sufficient representation of Definition 2.3. There, the index may be an information index, while the state may be taken as the model posterior for DMSO; the full history remains a fallback sufficient statistic.

Let $\mathcal{M}$ be the model class and $\mathcal{A}$ be the decision space. Model M has reward mean $r_M(a)$ and observation law $P_{M,a}$. For a candidate optimal action $a^\star$, write

$$\Delta_{M,a^\star}(p) := r_M(a^\star) - \mathbb{E}_{a\sim p} r_M(a).$$

AIR. An AIR belief is a pair belief $\nu \in \Delta(\mathcal{M}\times\mathcal{A})$ over $(M, A^\star)$, while the reference is an action marginal $q \in \operatorname{int}\Delta(\mathcal{A})$. Define

$$\mathrm{AIR}_{q,\eta}(p,\nu) := \mathbb{E}_{(M,A^\star)\sim\nu}\Delta_{M,A^\star}(p) - \frac{1}{\eta}\mathbb{E}_{a\sim p} I_\nu(A^\star; O \mid a) - \frac{1}{\eta} D_{\mathrm{KL}}(q_\nu \| q), \tag{12}$$

where q_ν is the $A^\star$-marginal of ν. Equivalently,

$$\mathrm{AIR}_{q,\eta}(p,\nu) = \mathbb{E}_{(M,A^\star),a,O}\left[\Delta_{M,A^\star}(\delta_a) - \frac{1}{\eta}\log\frac{\nu(A^\star \mid a, O)}{q(A^\star)}\right].$$

Lemma 3.4 (AIR gradient bracket)**.** *Assume finite spaces, or dominated spaces where the displayed Gateaux derivatives exist. Up to an additive constant on the simplex,*

$$\frac{\partial}{\partial\nu(M,a^\star)}\mathrm{AIR}_{q,\eta}(p,\nu) = \Delta_{M,a^\star}(p) - \frac{1}{\eta}\mathbb{E}_{a\sim p, O\sim P_{M,a}}\log\frac{\nu(A^\star = a^\star \mid a, O)}{q(a^\star)}. \tag{13}$$

Consequently, for every fixed pair $(M, a^\star)$,

$$\begin{aligned} B^{\mathrm{AIR}}_{q,\eta}(p,\nu; M, a^\star) &:= \mathrm{AIR}_{q,\eta}(p,\nu) + \left\langle \nabla_\nu \mathrm{AIR}_{q,\eta}(p,\nu), \delta_{(M,a^\star)} - \nu\right\rangle & (14)\\ &= \Delta_{M,a^\star}(p) - \frac{1}{\eta}\mathbb{E}_{a\sim p, O\sim P_{M,a}}\log\frac{\nu(A^\star = a^\star \mid a, O)}{q(a^\star)}. & (15)\end{aligned}$$

Proof. For fixed a, set

$$w_{a^\star,o} = \sum_M \nu(M,a^\star) P_{M,a}(o), \qquad w_o = \sum_{a^\star} w_{a^\star,o}.$$

Then

$$I_\nu(A^\star; O \mid a) + D_{\mathrm{KL}}(q_\nu\|q) = \sum_{a^\star,o} w_{a^\star,o}\log\frac{w_{a^\star,o}}{w_o q(a^\star)}.$$

Differentiating with respect to $\nu(M,a^\star)$ gives

$$\mathbb{E}_{O\sim P_{M,a}}\log\frac{\nu(A^\star = a^\star \mid a, O)}{q(a^\star)},$$

because the $+1$ terms from differentiating $w_{a^\star,o}\log w_{a^\star,o}$ and $w_o \log w_o$ cancel. Adding the derivative of the linear regret term proves (13). Pairing the gradient with $\delta_{(M,a^\star)} - \nu$ cancels the posterior-averaged terms and gives (14). $\square$

Theorem 3.5 (Exact AIR regret identity)**.** *Let an algorithm generate decision laws p_t, pair beliefs ν_t, and reference action beliefs $q_t \in \operatorname{int}\Delta(\mathcal{A})$. After action a_t and observation O_t, set*

$$q_{t+1}(\cdot) = \nu_t(A^\star \in \cdot \mid a_t, O_t).$$

Fix a model $M^\star$ and a comparator $a^\star$ with $q_t(a^\star) > 0$ for all relevant t; in the standard stochastic case, take $a^\star \in \arg\max_a r_{M^\star}(a)$. Let

$$\mathrm{Reg}_T(\mathsf{Alg}, M^\star, a^\star) := \mathbb{E}\sum_{t=1}^{T} \left(r_{M^\star}(a^\star) - r_{M^\star}(A_t)\right).$$

Then, for every $\eta > 0$,

$$\mathrm{Reg}_T(\mathsf{Alg}, M^\star, a^\star) = \frac{1}{\eta}\mathbb{E}\log\frac{q_{T+1}(a^\star)}{q_1(a^\star)} + \mathbb{E}\sum_{t=1}^{T} B^{\mathrm{AIR}}_{q_t,\eta}(p_t, \nu_t; M^\star, a^\star). \tag{16}$$

Remark 3.6 (Adversarial model sequence)**.** *Since Lemma 3.4 holds pointwise for every model–action pair, and does not require the true model to be stochastic, Theorem 3.5 extends directly to an adversarial model sequence $\mathbf{M} = (M_1, \ldots, M_T) \in \mathcal{M}^T$. In this case, the comparator is the best-in-hindsight action $a^\star$, and the regret identity remains exact. We omit this extension only to avoid additional notation.*

Proof. Condition on H_{t-1}. By the definition of the action-marginal posterior,

$$\mathbb{E}\left[\log\frac{q_{t+1}(a^\star)}{q_t(a^\star)}\middle| H_{t-1}\right] = \mathbb{E}_{a\sim p_t, O\sim P_{M^\star,a}}\log\frac{\nu_t(A^\star = a^\star \mid a, O)}{q_t(a^\star)}.$$

Combining this equality with (14) gives a one-step equality between regret, bracket, and posterior-ratio increment. Taking expectations and summing over time telescopes the logarithm. $\square$

MAIR. For MAIR the belief is $\mu \in \Delta(\mathcal{M})$ and the reference is $\rho \in \operatorname{int}\Delta(\mathcal{M})$. Write $\mu_{O|a} = \int P_{M,a}\mu(dM)$ and

$$\Delta_M(p) = r_M(a^\star_M) - \mathbb{E}_{a\sim p} r_M(a), \qquad a^\star_M \in \arg\max_a r_M(a).$$

Define

$$\mathrm{MAIR}_{\rho,\eta}(p,\mu) := \mathbb{E}_{M\sim\mu}\Delta_M(p) - \frac{1}{\eta}\mathbb{E}_{a\sim p} I_\mu(M; O \mid a) - \frac{1}{\eta}D_{\mathrm{KL}}(\mu\|\rho). \tag{17}$$

The same differentiation gives

$$\begin{aligned} B^{\mathrm{MAIR}}_{\rho,\eta}(p,\mu;M^\star) &:= \mathrm{MAIR}_{\rho,\eta}(p,\mu) + \langle\nabla_\mu \mathrm{MAIR}_{\rho,\eta}(p,\mu), \delta_{M^\star} - \mu\rangle && (18)\\ &= \Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{M^\star,a}\|\mu_{O|a}) - \frac{1}{\eta}\log\frac{\mu(M^\star)}{\rho(M^\star)}. && (19)\end{aligned}$$

If $\rho_{t+1}(\cdot) = \mu_t(\cdot \mid a_t, O_t)$, then

$$\mathrm{Reg}_T(\mathsf{Alg}, M^\star) = \frac{1}{\eta}\mathbb{E}\log\frac{\rho_{T+1}(M^\star)}{\rho_1(M^\star)} + \mathbb{E}\sum_{t=1}^{T} B^{\mathrm{MAIR}}_{\rho_t,\eta}(p_t, \mu_t; M^\star). \tag{20}$$

The proof is the same posterior-ratio telescope as for AIR. Thus AIR and MAIR are one indexed identity with different choices of Y.

Remark 3.7 (Danskin, Nash equilibrium, and why the bracket is not accidental)**.** *The direct proofs above are intentionally short. The original AIR proof of Xu and Zeevi (2025, Section 5) gives the variational reason behind the same equations. Introduce an auxiliary prediction rule $Q : (a, o) \mapsto \Delta(\mathcal{A})$ and the posterior-scoring objective*

$$\mathbb{E}_{(M,A^\star),a,O}\left[\Delta_{M,A^\star}(\delta_a) - \frac{1}{\eta}\log\frac{Q[a,O](A^\star)}{q(A^\star)}\right].$$

For fixed ν, the logarithmic score is strongly convex in the predictive distribution and is uniquely minimized by the Bayesian posterior $Q_\nu[a, o] = \nu(A^\star \in \cdot \mid a, o)$. The optimized value is the AIR objective; the unoptimized objective is linear, hence concave, in ν. Danskin's theorem identifies the AIR gradient with the derivative at the optimizer Q_ν. This is the conceptual content of Lemma 5.3 in Xu and Zeevi (2025). The same concavity plus strong convexity also gives the Nash equilibrium statement of Lemma 5.2 by first-order/Danskin reasoning, without invoking a separate fixed-point theorem. This does not necessarily simplify the full proof, because compactness, differentiability, and boundary conditions still have to be checked. It does explain why the gradient bracket is not an accidental cancellation: posterior scoring produces the minimizer, Danskin produces the gradient, and the regret identity telescopes the posterior reference.

The perspective developed in Xu and Zeevi (2025) is that frequentist sequential learning can be organized around algorithmic beliefs μ, interpreted as predictive laws, and decision rules induced by reference or posterior beliefs ρ, either through posterior sampling or through robust minimax optimization. We use MAIR to illustrate how a single functional captures both the confidence multiplier in UCB-type algorithms and the decision-estimation coefficient in one-round minimax rules.

Lemma 3.8 (MAIR gradient bracket and three canonical specializations)**.** *Assume the observation laws are dominated and the displayed derivatives exist. For fixed p, ρ, μ, up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial\mu(M)}\mathrm{MAIR}_{\rho,\eta}(p,\mu) = \Delta_M(p) - \frac{1}{\eta}\log\frac{\mu(M)}{\rho(M)} - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}D_{\mathrm{KL}}(P_{M,\pi}\|\mu_{o|\pi}).$$

Consequently, for every $M^\star$,

$$\begin{aligned}&\mathrm{MAIR}_{\rho,\eta}(p,\mu) + \langle\nabla_\mu\mathrm{MAIR}_{\rho,\eta}(p,\mu), \delta_{M^\star} - \mu\rangle\\ &\quad= \Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}D_{\mathrm{KL}}(P_{M^\star,\pi}\|\mu_{o|\pi}) - \frac{1}{\eta}\log\frac{\mu(M^\star)}{\rho(M^\star)}.\end{aligned} \tag{21}$$

This bracket has three important specializations.

1. *If $\mu = \rho$, the logarithmic correction vanishes and the bracket becomes the KL-DEC offset*

$$\Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}D_{\mathrm{KL}}(P_{M^\star,\pi}\|\mu_{o|\pi}). \tag{22}$$

   In Section 3.4 we will show that for GP-UCB,

   self-normalized calibration + UCB optimism $\Rightarrow$ MAIR gradient-error condition.

2. *If $\bar\mu \in \arg\max_{\mu\in\Delta(\mathcal{M})}\mathrm{MAIR}_{\rho,\eta}(p,\mu)$ is an interior maximizer, then*

$$\mathrm{MAIR}_{\rho,\eta}(p,\bar\mu) + \langle\nabla_\mu\mathrm{MAIR}_{\rho,\eta}(p,\bar\mu), \delta_{M^\star} - \bar\mu\rangle \le \mathrm{MAIR}_{\rho,\eta}(p,\bar\mu). \tag{23}$$

   This is the MAMS/MEBO bracket control.

3. Let $p_{M,\pi} = dP_{M,\pi}/d\nu_\pi$ and define, for an auxiliary pair $(\bar\pi, \bar o)$,

$$\frac{d\mu^{1/2}_{\bar\pi,\bar o}}{d\rho}(M) := \frac{\sqrt{p_{M,\bar\pi}(\bar o)}}{\int \sqrt{p_{N,\bar\pi}(\bar o)}\,\rho(dN)}. \tag{24}$$

Let $h^2(P,Q) := 1 - \int \sqrt{dP\,dQ}$. If $\bar\pi \sim p$ and $\bar o \sim P_{M^\star,\bar\pi}$, then

$$\begin{aligned}\mathbb{E}_{\bar\pi,\bar o}\Big[&\mathrm{MAIR}_{\rho,\eta}(p, \mu^{1/2}_{\bar\pi,\bar o}) + \langle \nabla_\mu \mathrm{MAIR}_{\rho,\eta}(p, \mu^{1/2}_{\bar\pi,\bar o}), \delta_{M^\star} - \mu^{1/2}_{\bar\pi,\bar o}\rangle\Big] \\ &\leq \Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}\mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\pi}, P_{M,\pi}). \end{aligned}\tag{25}$$

Thus the square-root posterior turns the MAIR bracket into a Hellinger E2D/DEC offset.

Proof. The derivative calculation is direct. The derivative of $\mathbb{E}_\mu \Delta_M(p)$ is $\Delta_M(p)$; the derivative of $D_{\mathrm{KL}}(\mu\|\rho)$ is $\log(\mu(M)/\rho(M)) + 1$; and differentiating the predictive mixture in the mutual-information term gives $D_{\mathrm{KL}}(P_{M,\pi}\|\mu_{o|\pi})$, up to the same simplex constant. Pairing the derivative with $\delta_{M^\star} - \mu$ cancels the averaged terms in (17) and proves (21).

The case $\mu = \rho$ is immediate from the logarithmic cancellation.

The case $\bar\mu \in \arg\max \mathrm{MAIR}$ follows from first-order optimality.

It remains to justify the square-root claim; this is a standard fact for the squared Hellinger distance. For fixed $\bar\pi$, let

$$Z_{\bar\pi}(\bar o) := \int \sqrt{p_{N,\bar\pi}(\bar o)}\,\rho(dN).$$

Then

$$\log \frac{d\mu^{1/2}_{\bar\pi,\bar o}}{d\rho}(M^\star) = \frac{1}{2}\log p_{M^\star,\bar\pi}(\bar o) - \log Z_{\bar\pi}(\bar o).$$

Under $\bar o \sim P_{M^\star,\bar\pi}$, Jensen's inequality gives

$$\begin{aligned}\mathbb{E}\log \frac{d\mu^{1/2}_{\bar\pi,\bar o}}{d\rho}(M^\star) &= -\mathbb{E}\log \frac{Z_{\bar\pi}(\bar o)}{\sqrt{p_{M^\star,\bar\pi}(\bar o)}} \\ &\geq -\log \int \sqrt{p_{M^\star,\bar\pi}(o)} Z_{\bar\pi}(o) d\nu_{\bar\pi}(o) \\ &= -\log\big(1 - \mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\bar\pi}, P_{M,\bar\pi})\big) \\ &\geq \mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\bar\pi}, P_{M,\bar\pi}).\end{aligned}$$

Averaging over $\bar\pi \sim p$, substituting into (21), and dropping the nonnegative KL term proves (25). ☐

3.4 Upper bounds: one identity and three algorithm families

We first present a generic indexed-information supersolution, in line with the philosophy of admissible relaxations in online learning (Rakhlin et al., 2012), but extended to partial-observation settings through information gain.

Let Φ_t be any predictable potential on states. For every algorithm, the algebraic identity

$$\begin{aligned}\mathbb{E}L_\Omega(H_T) = \mathbb{E}\sum_{t=1}^{T}\left[\ell(S_t, p_t) + \mathbb{E}[\Phi_{t+1}(S_{t+1}) \mid S_t, p_t] - \Phi_t(S_t)\right] \quad &(26) \\ + \Phi_1(s_1) - \mathbb{E}\Phi_{T+1}(S_{T+1}) \quad &(27)\end{aligned}$$

holds by telescoping.

Theorem 3.9 (Generic indexed AIR/MAIR upper certificate)**.** *Suppose an algorithm chooses p_t such that for all reached states,*

$$\ell(S_t, p_t) + \mathbb{E}[\Phi_{t+1}(S_{t+1}) \mid S_t, p_t] - \Phi_t(S_t) \leq \gamma_t \mathcal{I}_\chi(S_t, p_t) + \varepsilon_t. \tag{28}$$

Then

$$\mathbb{E}L_\Omega(H_T) \leq \Phi_1(s_1) - \mathbb{E}\Phi_{T+1}(S_{T+1}) + \mathbb{E}\sum_{t=1}^{T} \gamma_t \mathcal{I}_\chi(S_t, p_t) + \sum_{t=1}^{T} \varepsilon_t. \tag{29}$$

If $\gamma_t \leq \gamma$ and $\Phi_{T+1} \geq 0$, then

$$\mathbb{E}L_\Omega(H_T) \leq \Phi_1(s_1) + \gamma I_\mu(Y; H_T) + \sum_t \varepsilon_t.$$

Proof. Substitute (28) into (26). The last display uses Proposition 3.2. □

The bracket in (28) is the common object. UCB, E2D/DEC, and AMS/EBO differ in how they control it. In the model-index specialization, write the posterior/reference belief as μ_t and define the fixed-truth MAIR information increment

$$J_t^{\mathrm{MAIR}}(p; \omega^\star) := \mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{\omega^\star,t}(\cdot \mid S_t, a) \,\|\, P_{\mu_t,t}(\cdot \mid S_t, a)) \,. \tag{30}$$

The MAIR gradient bracket is the quantity obtained from the left-hand side of (28) after subtracting $\eta^{-1} J_t^{\mathrm{MAIR}}(p; \omega^\star)$, together with the posterior-ratio term in (20). The action-index AIR version is identical with the index $Y = A^\star$ and the conditional predictive laws $P_{t,y,a}$ in place of $P_{\omega,t,a}$. Thus the same theorem may be read either as an AIR statement about learning the optimal decision or as a MAIR statement about estimating the model/environment.

1. UCB families: calibration plus optimism. An upper-confidence-bound (UCB) algorithm maintains confidence sets or posterior predictive bands and chooses an action that is optimistic within that representation (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010). Calibration says the fixed truth lies inside the represented uncertainty set. Optimism converts regret into a posterior width. A self-normalized martingale inequality or elliptical-potential/log-determinant lemma then sums the squared widths into a realized information telescope. In this language, UCB does not solve the AIR/MAIR bracket optimization; it certifies the bracket after the action is chosen.

Lemma 3.10 (Self-normalized optimism implies bracket control)**.** *Fix a truth $\omega^\star$ and suppose that on an event $\mathcal{E}$, for every reached round t there exist a predictable width $w_t(a) \geq 0$, a tractable information increment $g_t \geq 0$, and constants $c_1, c_2 > 0$ such that*

$$\ell_{\omega^\star}(S_t, p_t) \leq c_1 \beta_t \left(\mathbb{E}_{a\sim p_t} w_t(a)^2\right)^{1/2}, \qquad \mathbb{E}_{a\sim p_t} w_t(a)^2 \leq c_2 g_t. \tag{31}$$

Then for every $\eta > 0$,

$$\ell_{\omega^\star}(S_t, p_t) - \frac{1}{\eta} g_t \leq \frac{\eta c_1^2 c_2 \beta_t^2}{4}. \tag{32}$$

If the exact fixed-truth indexed bracket uses $J_{\chi,t} := J_\chi(S_t, p_t; \omega^\star)$, then

$$B_{\chi,\eta}(S_t, p_t; \omega^\star) \leq \frac{\eta c_1^2 c_2 \beta_t^2}{4} + \frac{1}{\eta}(g_t - J_{\chi,t}). \tag{33}$$

Proof. On the event $\mathcal{E}$, by (31),

$$\ell_{\omega^\star}(S_t, p_t) \le c_1 \beta_t \sqrt{c_2 g_t}.$$

Young's inequality $ab \le \eta a^2/4 + b^2/\eta$ gives (32). Subtracting $\eta^{-1} J_{\chi,t}$ instead of $\eta^{-1} g_t$ gives (33). $\square$

The correction $g_t - J_{\chi,t}$ is not a second estimation cost. It is the algebraic difference between the fixed-truth posterior log gain in the gradient bracket and the tractable reference information increment. In linear-UCB and GP-UCB proofs this difference either cancels against the posterior-ratio telescope or is bounded by the same determinant/elliptical-potential term. Therefore

$$\text{self-normalized calibration} + \text{optimism} \quad \Longrightarrow \quad \text{AIR/MAIR gradient-bracket control.}$$

A central organizing perspective of this paper is that the UCB proof can be read as a MAIR decomposition. The round-heterogeneous summation of posterior widths, usually controlled by the elliptical-potential or log-determinant argument, plays the role of a telescoping estimation-complexity term: it is the accumulated algorithmic information or fixed-truth estimation loss. The confidence multiplier, by contrast, enters as a per-round upper bound on the MAIR/DEC bracket, i.e., on the coefficient that converts posterior information into regret. This interpretation may be a bit counterintuitive at first sight from the finite-action UCB proof viewpoint: what is often called the "estimation" difficulty in self-normalized martingale concentration becomes, in MAIR language, part of the regret-to-information coefficient; while the elliptical-potential summation becomes the estimation-complexity telescope. This reversal is precisely the model-based view: posterior widths quantify the "estimation complexity" of the underlying parameter, as explained by Lattimore (2023). The resulting unification is therefore not merely an algebraic rearrangement of known facts. It reveals a common round-heterogeneous posterior-telescoping structure while separating the distinct estimation and coefficient mechanisms behind UCB, E2D, and AMS/EBO.

2. E2D/DEC: robust one-step offset optimization. The decision-estimation coefficient (DEC) (Foster et al., 2021) and estimation-to-decision (E2D) algorithms choose p by optimizing a one-step Lagrangian relaxation. In model-index notation this takes the schematic form

$$\inf_{p \in \Delta(\mathcal{A})} \sup_{\omega \in \mathcal{N}(s)} \left\{ \ell_\omega(s, p) - \gamma \mathbb{E}_{a \sim p} D_{\mathrm{KL}}(P_{\omega,s,a} \| P_{\rho,s,a}) \right\}, \tag{34}$$

where ρ is a reference belief or reference model and $\mathcal{N}(s)$ is a localized comparison set that includes at least one environment generating the state s. The same one-round minimax optimization principle also applies to the original Hellinger form and constrained variants (Foster et al., 2023). Equation (34) is exactly the MAIR gradient bracket in (22) for the KL-DEC offset, or in (25) for the Hellinger DEC offset, but with the full Bellman information potential from (28),

$$\mathbb{E}\Phi_{t+1}(\tau(s, A, O)) - \Phi_t(s),$$

removed; here $\tau(s, A, O)$ is the posterior state notation as in Section 2.1.

Thus E2D/DEC is a one-step relaxation of the indexed Bellman program. It can be sharp when the posterior geometry is simple and the one-step witnesses paste together dynamically. Its role with realized information gain is subtler: an unlocalized DEC may be unbounded or vacuous, while a localized DEC must be compatible with the same posterior-reference telescope used by the upper or lower proof. Otherwise one risks combining a coefficient that already contains estimation difficulty with a separate log-determinant telescope.

3. AMS/EBO: robust Bellman information potential optimization. Adaptive minimax sampling, exploration-by-optimization, and mirror-descent information-ratio methods (Lattimore and György, 2021; Xu and Zeevi, 2025) can be viewed as robust controls of the indexed AIR bracket. AMS may be viewed as a constructive simplification of EBO: it operates directly over decision distributions and algorithmic beliefs, thereby avoiding the nonconstructive functional-estimator step. We now describe its close relationship with the Bellman information potential dynamic program in Theorem 3.9.

Select the information index

$$Y = \chi(\Omega).$$

At state s, let $q \in \operatorname{int}\Delta(\mathcal{Y})$ be the current reference distribution over Y, and let $\nu \in \mathcal{U}_t(s)$ be a local algorithmic belief over environments that supports at least one environment generating the state s. Write $P_{\nu,s,a} := \int P_{\omega,s,a}\nu(d\omega)$ and $\ell_\nu(s,p) := \int \ell_\omega(s,p)\nu(d\omega)$. Write

$$\nu_\chi := \mathcal{L}_\nu(Y), \qquad \nu_\chi^{a,o} := \mathcal{L}_\nu(Y \mid s,a,o),$$

where $\mathcal{L}$ is the conditional law notation as in Section 2. The AIR bracket can be written as a potential identity. Let V_t be an optional regret continuation potential, introduced here for illustration, and define the total Bellman information potential

$$\Phi_t^{q,\nu}(s) := V_t(s) + \gamma\,\mathbb{E}_{Y\sim\nu_\chi}[-\log q(Y)] = V_t(s) + \gamma H(\nu_\chi) + \gamma D_{\mathrm{KL}}(\nu_\chi \| q).$$

If, after action a and observation o, the belief is updated to $\nu^{a,o}$ and the next reference is chosen as $q^+ = \nu_\chi^{a,o}$, then

$$\begin{aligned}
&\mathbb{E}_{\substack{a\sim p\\ o\sim P_{\nu,s,a}}} \Phi_{t+1}^{q^+,\nu^{a,o}}(S') - \Phi_t^{q,\nu}(s)\\
&\qquad = \mathbb{E}V_{t+1}(S') - V_t(s) - \gamma\,\mathcal{I}_\chi(s,p;\nu) - \gamma D_{\mathrm{KL}}(\nu_\chi\|q).
\end{aligned}$$

Hence the robust EBO decision rule, after the exchange-of-decision simplification of Xu and Zeevi (2025), may be written schematically as

$$\inf_{p\in\Delta(\mathcal{A})}\sup_{\nu\in\mathcal{U}_t(s)}\left\{\ell_\nu(s,p) + \mathbb{E}\Phi_{t+1}^{\nu_\chi^{a,O},\nu^{a,O}}(S') - \Phi_t^{q_t,\nu}(s)\right\}.$$

Equivalently, in AIR-bracket form, this gives the Bellman extension of AMS:

$$\inf_{p\in\Delta(\mathcal{A}_s)}\sup_{\nu\in\mathcal{U}_t(s,q_t)}\left\{\ell_\nu(s,p) + \mathbb{E}V_{t+1}(S') - V_t(s) - \gamma\,\mathcal{I}_\chi(s,p;\nu) - \gamma D_{\mathrm{KL}}(\nu_\chi\|q_t)\right\}.$$

The standard AMS/EBO bracket is recovered by taking $V_t \equiv 0$; the display above gives its Bellman extension, which may be useful for non-episodic formulations and computational considerations.

The corresponding reference update is

$$q_{t+1} = (\nu_t)_\chi^{a_t,o_t},$$

where ν_t is the algorithmic belief selected by the saddle problem, $a_t \sim p_t$, and o_t is the observed feedback. Thus the reference itself evolves, and the regularized information terms telescope as log gains: for a dynamically consistent reference process and a fixed indexed truth $Y^\star$ with $q_1(Y^\star) > 0$,

$$\sum_{t=1}^{T}\log\frac{q_{t+1}(Y^\star)}{q_t(Y^\star)} = \log\frac{q_{T+1}(Y^\star)}{q_1(Y^\star)} \le \log\frac{1}{q_1(Y^\star)}.$$

Consequently, if the robust AIR bracket is bounded by ϵ_t along the trajectory and the algorithmic beliefs are calibrated to the fixed truth, then

$$\mathbb{E}\,\mathrm{Reg}_T \le V_1(s_1) - \mathbb{E}V_{T+1}(S_{T+1}) + \gamma \log \frac{1}{q_1(Y^\star)} + \sum_{t=1}^{T} \epsilon_t + \mathrm{cal}_T,$$

where cal_T denotes the indexed AIR calibration error; under a Bayes prior, the displayed logarithm is replaced by $\mathbb{E}[-\log q_1(Y)]$. This is the sense in which AMS/EBO need not rely on a full model-based specification: its complexity can scale with the information index rather than with the ambient model dimension.

Remark 3.11 (The one identity)**.** *The identity is the potential telescope (26), refined by the AIR/MAIR posterior-ratio identities. UCB certifies the bracket through confidence and optimism. E2D/DEC greedily optimizes the MAIR/DEC offset. AMS/EBO controls a robust saddle over indexed beliefs. The algorithms differ, but they act on the same algebraic decomposition of regret into a potential telescope plus an indexed-information bracket.*

4 Lower bounds: reference histories and quantile indices

4.1 Ghost probability and true good probability

Fix an average-regret threshold $r \ge 0$. The posterior-reference ghost-good probability is

$$p_r^{\mathsf{Alg}}(\mu,\chi) := \mathbb{P}_{Y\sim\mu_\chi,\, H_T'\sim\bar{\mathbb{P}}_\mu^{\mathsf{Alg}}}\left(\bar{L}_\chi(Y,H_T') \le r\right), \tag{35}$$

where Y and H_T' are independent under the ghost law and μ_χ is the marginal law of Y. The true good probability is

$$q_r^{\mathsf{Alg}}(\mu,\chi) := \mathbb{P}_{Y,H_T}\left(\bar{L}_\chi(Y,H_T) \le r\right), \tag{36}$$

where (Y,H_T) are generated by $\Omega\sim\mu$ and $H_T\sim\mathbb{P}_\Omega^{\mathsf{Alg}}$.

The ghost probability is algorithm-dependent. That dependence is intentional: it preserves the reference trajectory geometry instead of replacing it with a worst-case static small ball.

4.2 Reference-history quantile theorem

Let

$$\mathrm{kl}(q,p) = q\log\frac{q}{p} + (1-q)\log\frac{1-q}{1-p}$$

be binary KL divergence. For $p\in[0,1]$ and $c\ge 0$, define

$$\psi(p,c) := \sup\{q\in[0,1] : \mathrm{kl}(q,p)\le c\}.$$

Theorem 4.1 (Reference-history quantile indexed-information lower bound)**.** *For every prior μ, index map χ, threshold $r\ge 0$, and algorithm* Alg*,*

$$\mathrm{kl}\big(q_r^{\mathsf{Alg}}(\mu,\chi), p_r^{\mathsf{Alg}}(\mu,\chi)\big) \le C_\chi^{\mathsf{Alg}}(\mu) = T\bar{C}_\chi^{\mathsf{Alg}}(\mu). \tag{37}$$

Consequently,

$$\mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \ge Tr\left[1-\psi\big(p_r^{\mathsf{Alg}}(\mu,\chi), T\bar{C}_\chi^{\mathsf{Alg}}(\mu)\big)\right]. \tag{38}$$

In particular, if

$$T\bar{C}_{\chi}^{\mathsf{Alg}}(\mu) \le \mathrm{kl}\left(\frac{1}{2}, p_r^{\mathsf{Alg}}(\mu,\chi)\right), \tag{39}$$

then

$$\mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_{\Omega}^{\mathsf{Alg}}}\bar{L}_{\Omega}(H_T) \ge \frac{r}{2}, \qquad \mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_{\Omega}^{\mathsf{Alg}}}L_{\Omega}(H_T) \ge \frac{Tr}{2}. \tag{40}$$

A convenient sufficient condition is

$$T\bar{C}_{\chi}^{\mathsf{Alg}}(\mu) + \log 2 \le \frac{1}{2}\log\frac{1}{p_r^{\mathsf{Alg}}(\mu,\chi)}. \tag{41}$$

Proof. Consider the true joint law of (Y, H_T) and the ghost law of (Y, H_T'). Both have the same marginal law of Y, while Y and H_T' are independent under the ghost law. Hence

$$D_{\mathrm{KL}}\big(\mathcal{L}(Y,H_T)\|\mathcal{L}(Y,H_T')\big) = I_\mu(Y;H_T).$$

Data processing applied to the indicator of the event $\{\bar{L}_\chi(Y,\cdot) \le r\}$ gives

$$\mathrm{kl}(q_r^{\mathsf{Alg}}, p_r^{\mathsf{Alg}}) \le I_\mu(Y;H_T).$$

Proposition 3.2 identifies the right-hand side with $C_\chi^{\mathsf{Alg}}(\mu) = T\bar{C}_\chi^{\mathsf{Alg}}(\mu)$, proving (37). The definition of ψ gives $q_r^{\mathsf{Alg}} \le \psi(p_r^{\mathsf{Alg}}, T\bar{C}_\chi^{\mathsf{Alg}})$. By nonnegativity of loss and by (3),

$$\begin{aligned}\mathbb{E}_{\Omega,H_T}L_\Omega(H_T) &= \mathbb{E}_{Y,H_T}L_\chi(Y,H_T)\\ &\ge Tr\,\mathbb{P}\{\bar{L}_\chi(Y,H_T) > r\}\\ &= Tr(1-q_r^{\mathsf{Alg}}),\end{aligned}$$

which proves (38). If (39) holds, then $q_r^{\mathsf{Alg}} \le 1/2$, giving (40). Finally,

$$\mathrm{kl}\left(\frac{1}{2}, p\right) = \frac{1}{2}\log\frac{1}{4p(1-p)} \ge \frac{1}{2}\log\frac{1}{p} - \log 2,$$

so (41) implies (39). ☐

Relation to interactive Fano. Theorem 4.1 should be viewed as an indexed, posterior-reference specialization of the interactive Fano method of Chen et al. (2024). The binary-KL step is the same ghost-data argument: one compares the true experiment with a reference experiment, applies data processing to the small-regret event, and obtains a Bayes risk lower bound when the trajectory information is smaller than the logarithmic inverse ghost-good probability. The additional point here is representational. We choose the reference experiment to be the Bayesian posterior-predictive trajectory and choose an information index $Y = \chi(\Omega)$. With this choice, the trajectory information admits the exact stepwise identity

$$I_\mu(Y;H_T) = \mathbb{E}_{H_T'\sim\bar{\mathbb{P}}_\mu^{\mathsf{Alg}}}\sum_{t=1}^{T}\mathcal{I}_\chi(S_t', p_t').$$

Thus the same reference history H_T' generates both the ghost quantile and the indexed information telescope. This is the lower-bound counterpart of the AIR/MAIR upper-bound identity. In particular, $\chi(\Omega) = \Omega$ gives the model-index MAIR form, while $\chi(\Omega) = \pi^\star(\Omega)$ gives the action-index AIR form.

The novelty is not a new Fano inequality, but a new representation of interactive Fano in posterior-reference indexed-information coordinates. This avoids prematurely collapsing the adaptive experiment to a one-round decision distribution, the step that creates the localized change-of-measure difficulty in one-round DEC reductions. It also makes the lower-bound critical radius directly comparable to AIR/MAIR supersolution upper bounds, since both sides are expressed as sums of the same indexed information along posterior-reference histories.

Remark 4.2 (Why average notation is useful)**.** *The theorem could be written entirely with cumulative quantities. Writing $\bar{C}_\chi = T^{-1}\mathbb{E}\sum_t \mathcal{I}_\chi$ and $\bar{L} = T^{-1}L$ makes the scaling visible:*

$$\textit{information side} = T\bar{C}_\chi, \qquad \textit{regret side} = Tr.$$

Thus T appears exactly as in one-round reductions, but no posterior trajectory has been collapsed into a static action distribution.

4.3 Bellman-Fano lower certificates

Theorem 4.1 is algorithm-specific. We now give algorithm-uniform certificates. The first bounds the information of any algorithm; the second bounds the ghost probability of success of any algorithm.

Information supersolution. For a function F on states, define

$$(\mathsf{C}_t F)(s) := \sup_{p\in\Delta(\mathcal{A}(s))} \left\{ \mathcal{I}_\chi(s,p) + \mathbb{E}_{a\sim p, o\sim P_{s,a}} F(\tau(s,a,o)) \right\}.$$

A sequence $\overline{C}_t$ is an information supersolution if

$$\overline{C}_{T+1} \geq 0, \qquad \overline{C}_t(s) \geq (\mathsf{C}_t \overline{C}_{t+1})(s) \quad \forall t, s.$$

Then, for every algorithm starting from s_1,

$$C_\chi^{\mathsf{Alg}}(\mu) \leq \overline{C}_1(s_1). \tag{42}$$

The proof is the same backward induction as Proposition 2.2.

Ghost-good-mass supersolution. Let $m \in \Delta(\mathcal{Y})$ be the target-index distribution used to evaluate success, let s be the posterior-reference state, and let $g : \mathcal{Y} \to [0, L]$ be an accumulated average-loss profile. For threshold r, define the exact recursion

$$\Gamma^r_{T+1}(m,s,g) = m\{y : g(y) \leq r\},$$

$$\Gamma^r_t(m,s,g) = \sup_{p\in\Delta(\mathcal{A}(s))} \mathbb{E}_{a\sim p, o\sim P_{s,a}} \Gamma^r_{t+1}\left(m, \tau(s,a,o), g + \frac{1}{T}\ell^\chi_\cdot(s,a)\right),$$

where $\ell^\chi_\cdot(s,a)$ is the function $y \mapsto \ell^\chi_y(s,a)$. If $\overline{\Gamma}^r_t$ is a supersolution of this recursion, then for every algorithm,

$$p_r^{\mathsf{Alg}}(\mu,\chi) \leq \overline{\Gamma}^r_1(\mu_\chi, s_1, 0). \tag{43}$$

Define the ghost entropy lower certificate

$$\underline{\mathcal{E}}^r_1(\mu,\chi) := -\log \overline{\Gamma}^r_1(\mu_\chi, s_1, 0). \tag{44}$$

Theorem 4.3 (Bellman quantile-index lower certificate)**.** *Suppose $\overline{C}_t$ is an information supersolution and $\overline{\Gamma}_t^r$ is a ghost-good-mass supersolution. Then*

$$\inf_{\mathsf{Alg}} \mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \geq Tr\left[1-\psi\left(e^{-\underline{\mathcal{E}}_1^r(\mu,\chi)}, \overline{C}_1(s_1)\right)\right]. \tag{45}$$

In particular, if

$$\overline{C}_1(s_1) + \log 2 \leq \frac{1}{2}\underline{\mathcal{E}}_1^r(\mu,\chi), \tag{46}$$

then

$$\inf_{\mathsf{Alg}} \mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}. \tag{47}$$

Consequently,

$$\mathfrak{R}_T^\star \geq \sup_{\mu,\chi,r}\left\{\frac{Tr}{2} : \overline{C}_1(s_1) + \log 2 \leq \frac{1}{2}\underline{\mathcal{E}}_1^r(\mu,\chi)\right\}.$$

Proof. For every algorithm, (42) gives $T\bar{C}_\chi^{\mathsf{Alg}}(\mu) = C_\chi^{\mathsf{Alg}}(\mu) \leq \overline{C}_1(s_1)$, and (43) gives $p_r^{\mathsf{Alg}}(\mu,\chi) \leq e^{-\underline{\mathcal{E}}_1^r(\mu,\chi)}$. Substitute these two inequalities into Theorem 4.1, using the monotonicity of ψ in both arguments. The minimax statement follows from Yao's principle. ◻

The critical average-regret radius is

$$r_\star(\mu,\chi) := \sup\left\{r : \overline{C}_1(s_1) + \log 2 \leq \frac{1}{2}\underline{\mathcal{E}}_1^r(\mu,\chi)\right\}. \tag{48}$$

The lower bound is order $Tr_\star$. A two-point Le Cam proof corresponds to the special case in which $\underline{\mathcal{E}} = O(1)$. A Fano or local-prior-mass lower bound has $\underline{\mathcal{E}}$ of order dimension or effective dimension.

4.4 DEC as one-step relaxation

As discussed in Section 3.4, offset DEC is obtained by taking the indexed Bellman upper bracket in (28) and making a one-step relaxation: the future-value term is either dropped or absorbed into the instantaneous loss, and the posterior-reference law is replaced by a fixed one-step reference predictive law. In the model-index case $\chi(\omega) = \omega$, let $\bar{\nu}$ be a reference belief at state s, and write

$$P_{\bar{\nu},s,a} := \int P_{\omega,s,a}\,\bar{\nu}(d\omega)$$

for its predictive law. The corresponding pointwise KL offset is

$$\mathrm{dec}_\gamma^{\mathrm{KL}}(s;\bar{\nu}) := \inf_{p\in\Delta(\mathcal{A}_s)}\sup_{\omega\in\Omega_s}\left\{\mathbb{E}_{a\sim p}\ell_\omega(s,a) - \gamma\mathbb{E}_{a\sim p}D_{\mathrm{KL}}(P_{\omega,s,a}\,\|\,P_{\bar{\nu},s,a})\right\}. \tag{one-step model-index offset}$$

Thus the usual offset DEC is a worst-case, static-reference relaxation of the Bellman program: the learner chooses the one-step decision distribution p, and the local adversary then chooses the hardest model relative to the fixed reference predictive law.

The same one-step logic also appears on the lower-bound side, but only after an additional reduction from a dynamic posterior-reference experiment to a one-step reference experiment. Theorem 4.1 and Theorem 4.3 retain the dynamic reference posteriors. A one-step quantile DEC is obtained only by relaxing this dynamic proof: one fixes a reference model, or more generally a reference predictive law, decomposes the trajectory divergence into one-step divergences, and replaces the algorithm-dependent ghost history by a one-round decision distribution. In Chen et al.

(2024, Section 3.2.2), this idea is made explicit through the quantile PAC-DEC. A reference model $\bar{M}$ generates the ghost experiment, an adversarial model M supplies the hard alternative, and the loss certificate is a quantile of the one-step risk rather than its expectation. The further reduction to constrained regret-DEC in that paper is precisely where localization, regularity, and problem-specific assumptions enter.

The passage from the dynamic theorem to a one-step coefficient is therefore not merely notational. On both the upper- and lower-bound sides, existing DEC measures can be viewed as one-step relaxations obtained by freezing the posterior-reference process into reference models or predictive laws. This interpretation is primarily conceptual: even the distinction between quantile and expected loss raises nontrivial technical questions when deriving regret-DEC lower bounds under regularity conditions. Nevertheless, the present paper adopts this viewpoint because it clarifies how one-step DEC certificates arise from the dynamic indexed Bellman picture.

Thus DEC/E2D should be read as a one-step relaxation of indexed Bellman information complexity, rather than as a universally tight conversion coefficient from information to regret. When the posterior geometry is simple, this relaxation can be sharp. In more structured problems, however, a one-round coefficient can obscure the dynamic object in two distinct ways: it may charge an estimation or entropy cost that the posterior-reference lower bound has already paid, or it may discard the geometry of the reference trajectory that enables localization. The examples in Section 5 illustrate indexed-Bellman certificates and diagnostics; the following informal diagnostics explain why the corresponding one-round DEC accounting should be interpreted with care.

- *Vacuity for RKHS classes without localization.* In infinite-dimensional kernel bandits, a global one-round DEC over an RKHS ball can be unbounded or vacuous unless the comparison is localized to the finite active experiment actually visited by the algorithm; see the end of Section 5.1 for details. Known routes to matching DEC-style lower bounds introduce this localization explicitly or impose regularity conditions that enforce it implicitly through a critical-radius or sub-root fixed point (Foster et al., 2023). Indexed Bellman information complexity keeps this localization inside the posterior-reference process itself.

- *Double counting in multi-armed bandits.* The hard prior and the ghost entropy already contain the cost of identifying one arm among K. Therefore, after a Fano or posterior-reference calculation has produced the logarithmic entropy term, appending an independent DEC "estimation" penalty charges the same information budget a second time. This is not a statement that DEC-based algorithms are necessarily suboptimal for multi-armed bandits; it is a statement that the entropy appearing in the lower certificate and the estimation term appearing in a one-step upper relaxation should not be added as independent costs.

- *Collapsed lower bounds in linear bandits.* A one-round coefficient obtained by collapsing the adaptive trajectory to a single action distribution can leave a localization or logarithmic gap in the lower bound. The gap is not intrinsic to the linear-bandit experiment: it is introduced by replacing the posterior trajectory by a static action law. The indexed Bellman formulation keeps the posterior covariance, the information telescope, and the ghost quantile on the same reference path.

- *Representational ambiguity for Bellman rank.* Bellman rank is a representation-relative algebraic certificate: it depends on a hypothesis class, value-function class, roll-in distribution, discrepancy, and Bellman-error witness (Jiang et al., 2017). Without fixing these observable witnesses, bounded Bellman rank is not itself a DMSO model class and does not determine an intrinsic DEC. Once a bilinear-class witness and its estimation complexity are specified, DEC-style questions become meaningful (Du et al., 2021). Thus, a lower bound requires an explicit hard family, observable witnesses, and a ghost-entropy calculation within the specified class.

Definition 4.4 (Tight indexed Bellman representation)**.** *Fix a problem family, a horizon sequence, and a class of admissible indices χ. An indexed Bellman representation is* tight *if, for the relevant initial states s_1 and for some prior-index pair (μ, χ), there exist a regret supersolution $\overline{V}$ and Bellman-Fano lower certificates $\overline{C}, \underline{\mathcal{E}}$ on the same state space such that, up to the stated universal constants or logarithmic factors,*

$$\overline{V}_1(s_1) \lesssim T r_\star(\mu, \chi), \qquad \mathfrak{R}_T^\star \gtrsim T r_\star(\mu, \chi),$$

where $r_\star(\mu, \chi)$ is the critical average-regret radius in (48).

This definition is the representation-level replacement for asking whether a one-round DEC coefficient is universally tight. The question is not whether a single static coefficient is always exact, but whether the same Bellman-sufficient state supports both an upper supersolution and a posterior-reference quantile lower certificate at the same critical radius.

5 Applications and examples

Before turning to examples, we spell out what the state means in common problems. In DMSO, the state may be the posterior over models, with the full history as a fallback sufficient statistic. In multi-armed bandits, the state can be the vector of posterior sufficient statistics for arm means, and the natural AIR index is the optimal arm. In linear bandits, a Gaussian or confidence-state representation consists of a posterior mean and covariance, or their frequentist confidence analogues; the index may be the parameter, an optimal action, or a coarse optimal direction. In finite-action kernel bandits, the state is the active marginal posterior on the finite action set; the unknown function remains an RKHS element. In low-Bellman-rank problems, the state must include the witness features and value-function information needed to predict Bellman errors. Thus "state" never means a universal finite-dimensional object. It means the smallest representation, exact or approximate, on which the Bellman, information, and ghost-good recursions close.

5.1 Finite-action kernel bandits: GP–UCB, GP–E2D, and AMS/EBO

This subsection specializes the indexed Bellman language to kernel bandits on a finite active action set. The point is not to make the unknown function finite-dimensional. The truth remains an element of an RKHS. What becomes finite is the marginal experiment observed through the active actions. This finite marginal is enough to define the GP posterior update, the model-index MAIR information, and the action-index AIR belief over the optimal active action.

Problem and finite active marginal. Let $\mathcal{H}_k$ be a reproducing kernel Hilbert space (RKHS) on a possibly infinite domain $\mathcal{X}_{\text{all}}$, with kernel k satisfying $k(x,x) \leq \kappa^2$. The learner is evaluated on a finite active set

$$\mathcal{X} = \{x_1, \ldots, x_n\} \subset \mathcal{X}_{\text{all}}.$$

The unknown reward function $f^\star \in \mathcal{H}_k$ satisfies $\|f^\star\|_{\mathcal{H}_k} \leq B$, and at round t the learner chooses $X_t \in \mathcal{X}$ and observes

$$Y_t = f^\star(X_t) + \varepsilon_t, \qquad \mathbb{E}[\varepsilon_t \mid H_{t-1}, X_t] = 0, \tag{49}$$

where the noise is conditionally R-sub-Gaussian. For the Gaussian calculations below we use the algorithmic reference model with variance parameter $\lambda > 0$; the frequentist confidence event is obtained by the usual self-normalized argument.

Let $F = (f(x_1), \dots, f(x_n)) \in \mathbb{R}^n$ be the active reward vector and let $K_{\mathcal{X}}$ be the $n \times n$ kernel matrix. The GP reference prior is

$$F \sim N(0, K_{\mathcal{X}}), \qquad Y_t \mid F, X_t = x_i \sim N(F_i, \lambda).$$

After history H_{t-1}, the algorithmic posterior on F is Gaussian,

$$F \mid H_{t-1} \sim N(m_t, \Sigma_t).$$

For $x_i \in \mathcal{X}$, write $m_t(x_i) = e_i^\top m_t$ and $s_t^2(x_i) = e_i^\top \Sigma_t e_i$. If $X_t = x_i$, the exact posterior update is

$$m_{t+1} = m_t + \frac{\Sigma_t e_i}{\lambda + s_t^2(x_i)} \{Y_t - m_t(x_i)\}, \tag{50}$$

$$\Sigma_{t+1} = \Sigma_t - \frac{\Sigma_t e_i e_i^\top \Sigma_t}{\lambda + s_t^2(x_i)}. \tag{51}$$

This is a finite marginal posterior update. It should not be read as an assumption that $f^\star$ has n parameters. The RKHS function may be infinite-dimensional; only the observations and decisions in this finite experiment depend on the vector F.

Model-index information and action-index information. For the model-index specialization, take the index to be the active reward vector $Y_{\mathrm{MAIR}} = F$. The one-step GP/MAIR information at state H_{t-1} is

$$\mathcal{I}_t^{\mathrm{GP}}(p) := \mathbb{E}_{x \sim p} \frac{1}{2} \log\left(1 + \frac{s_t^2(x)}{\lambda}\right). \tag{52}$$

Along a realized action sequence,

$$C_T^{\mathrm{GP}} := \sum_{t=1}^{T} \mathcal{I}_t^{\mathrm{GP}}(\delta_{X_t}) = \frac{1}{2} \log\det\left(I + \lambda^{-1} K_{X_{1:T}, X_{1:T}}\right), \tag{53}$$

where repeated actions are included in the Gram matrix. This is the realized information gain of the algorithmic trajectory. It can be much smaller than the maximal information gain that maximizes over all length-T designs.

For the action-index specialization, assume ties are broken by a fixed rule and set

$$A^\star(F) \in \arg\max_{x \in \mathcal{X}} F_x, \qquad q_t(a) := \mathbb{P}(A^\star = a \mid H_{t-1}).$$

Let ν_t^a be the exact conditional law of F given H_{t-1} and $A^\star = a$. It is generally a truncated Gaussian on the cone where a is optimal. Define the conditional predictive law

$$P_{t,a,x}(\cdot) := \int N(F_x, \lambda)(\cdot)\, \nu_t^a(dF),$$

for the observation at action x given the event $A^\star = a$. The marginal predictive is

$$P_{t,x} = \sum_{a \in \mathcal{X}} q_t(a) P_{t,a,x} = N(m_t(x), \lambda + s_t^2(x)).$$

The AIR information increment is

$$\mathcal{I}_t^{\mathrm{AIR}}(p) := \mathbb{E}_{x\sim p} \sum_{a\in\mathcal{X}} q_t(a) D_{\mathrm{KL}}(P_{t,a,x} \| P_{t,x}). \tag{54}$$

The conditional AIR regret is

$$\Delta_t^{\mathrm{AIR}}(p) := \sum_{a\in\mathcal{X}} q_t(a) \mathbb{E}_{F\sim\nu_t^a} \mathbb{E}_{x\sim p}[F_a - F_x]. \tag{55}$$

The exact indexed chain rule gives

$$\sum_{t=1}^{T} \mathbb{E}\mathcal{I}_t^{\mathrm{AIR}}(p_t) = I(A^\star; H_T) \le H(A^\star) \le \log n.$$

Thus AIR places the entropy telescope on the decision index rather than on the ambient RKHS. The price is that the coefficient converting action-index information into regret must be controlled by an algorithmic bracket.

Algorithm 1: GP–UCB. GP–UCB uses the Gaussian marginal posterior (m_t, Σ_t) and chooses

$$X_t \in \arg\max_{x\in\mathcal{X}} \{m_t(x) + \beta_t s_t(x)\}. \tag{56}$$

The following event is the frequentist calibration certificate:

$$\mathcal{E}_{\mathrm{GP}} := \{|f^\star(x) - m_t(x)| \le \beta_t s_t(x) \quad \text{for all } t \le T, x \in \mathcal{X}\}. \tag{57}$$

For finite $\mathcal{X}$, standard RKHS self-normalized concentration gives $\mathbb{P}(\mathcal{E}_{\mathrm{GP}}) \ge 1 - \delta$ for the usual choice of β_t, for example

$$\beta_t \asymp B + R\sqrt{\log\det(I + \lambda^{-1} K_{\mathcal{X},t}) + \log(1/\delta)},$$

with the precise form depending on the normalization of k and the ridge parameter. The next theorem isolates the deterministic part of the GP–UCB proof.

Theorem 5.1 (GP–UCB realized-information bound)**.** *On the event $\mathcal{E}_{\mathrm{GP}}$, the GP–UCB rule (56) satisfies*

$$\mathrm{Reg}_T(f^\star) \le 2\beta_T \sum_{t=1}^{T} s_t(X_t) \le 2\beta_T \sqrt{T\, c_{\kappa,\lambda}\, C_T^{\mathrm{GP}}}, \tag{58}$$

where

$$c_{\kappa,\lambda} := \frac{2\kappa^2}{\log(1 + \kappa^2/\lambda)}$$

and C_T^{GP} is the realized information gain in (53). Consequently, if rewards are bounded in $[0,1]$ and $\mathbb{P}(\mathcal{E}_{\mathrm{GP}}) \ge 1 - \delta$, then

$$\mathbb{E}\,\mathrm{Reg}_T(f^\star) \le 2\beta_T \sqrt{T\, c_{\kappa,\lambda}\, \mathbb{E}C_T^{\mathrm{GP}}} + T\delta.$$

Proof. Let $x^\star \in \arg\max_{x\in\mathcal{X}} f^\star(x)$. On $\mathcal{E}_{\mathrm{GP}}$, optimism gives

$$f^\star(x^\star) \le m_t(x^\star) + \beta_t s_t(x^\star) \le m_t(X_t) + \beta_t s_t(X_t).$$

Calibration at X_t gives $m_t(X_t) \le f^\star(X_t) + \beta_t s_t(X_t)$. Hence

$$f^\star(x^\star) - f^\star(X_t) \le 2\beta_t s_t(X_t) \le 2\beta_T s_t(X_t).$$

Summing and applying Cauchy–Schwarz yields

$$\mathrm{Reg}_T(f^\star) \le 2\beta_T \sqrt{T \sum_{t=1}^{T} s_t^2(X_t)}.$$

For $u \in [0, \kappa^2/\lambda]$, monotonicity of $u/\log(1+u)$ on a bounded interval gives

$$\lambda u \le \frac{\kappa^2}{\log(1+\kappa^2/\lambda)} \log(1+u).$$

Applying this with $u = s_t^2(X_t)/\lambda$ and using (53) proves (58). The expectation bound adds the trivial regret bound T on $\mathcal{E}_{\mathrm{GP}}^c$ and uses Jensen's inequality. □

GP–UCB as a MAIR decomposition. The theorem is exactly Lemma 3.10 with the model-index information surrogate $g_t = \frac{1}{2}\log(1 + s_t^2(X_t)/\lambda)$. The log determinant $\sum_t g_t = C_T^{\mathrm{GP}}$ is the estimation-complexity telescope. The confidence multiplier β_t is the coefficient controlling the MAIR bracket. This is why GP–UCB is often practically strong: it follows the realized trajectory and pays the realized information gain, not the worst possible design information, even though its coefficient may be conservative in step-uniform minimax analyses.

Algorithm 2: GP–E2D. A GP–E2D rule keeps the same GP posterior reference but chooses a distribution over actions by solving a localized one-round MAIR/DEC offset. Given a localized comparison set $\mathcal{F}_t$ containing plausible functions, define

$$\Delta_f(p) := \max_{x\in\mathcal{X}} f(x) - \mathbb{E}_{x\sim p} f(x),$$

and let $P_{f,x} = N(f(x), \lambda)$ while $P_{t,x} = N(m_t(x), \lambda + s_t^2(x))$ is the posterior predictive. A schematic GP–E2D decision is

$$p_t \in \arg\min_{p\in\Delta(\mathcal{X})} \sup_{f\in\mathcal{F}_t} \left\{\Delta_f(p) - \gamma_t \mathbb{E}_{x\sim p} D_{\mathrm{KL}}(P_{f,x} \| P_{t,x})\right\}. \tag{59}$$

If $f^\star \in \mathcal{F}_t$ and the optimized value in (59) is at most ε_t, then

$$\Delta_{f^\star}(p_t) \le \gamma_t \mathbb{E}_{x\sim p_t} D_{\mathrm{KL}}(P_{f^\star,x} \| P_{t,x}) + \varepsilon_t. \tag{60}$$

Summing (60) gives a regret bound once the fixed-truth KL terms are related to a posterior-ratio or information telescope. Under a well-specified Bayesian GP reference this relation holds in expectation by the MAIR chain rule. Under a fixed frequentist truth it requires calibration/localization. This is the technical point that a GP–E2D analysis must address: the one-round DEC is optimized, but the compatibility between that coefficient and the realized GP log determinant is not automatic. An unlocalized offset over an infinite-dimensional RKHS ball can be unbounded or vacuous; meaningful GP–E2D guarantees must localize the comparison set to the posterior-reference experiment.

Algorithm 3: AMS/EBO on the action index. AMS/EBO attacks the same bracket from the robust posterior side. In the finite-action kernel marginal, the natural AIR index is $A^\star$. A robust AIR decision may be written schematically as

$$p_t \in \arg\min_{p\in\Delta(\mathcal{X})} \sup_{b\in\mathfrak{B}_t} \left\{\Delta_b^{\mathrm{AIR}}(p) - \gamma_t \mathcal{I}_b^{\mathrm{AIR}}(p)\right\}, \tag{61}$$

where $\mathfrak{B}_t$ is a set of posterior/action-index beliefs around the current reference belief. If the chosen p_t satisfies

$$\Delta_t^{\mathrm{AIR}}(p_t) \le \gamma_t \mathcal{I}_t^{\mathrm{AIR}}(p_t) + \varepsilon_t,$$

then the Bayes regret under the algorithmic reference is bounded by

$$\mathbb{E}\,\mathrm{Reg}_T \le \sum_{t=1}^T \gamma_t \mathbb{E}\mathcal{I}_t^{\mathrm{AIR}}(p_t) + \sum_{t=1}^T \varepsilon_t \le \gamma H(A^\star) + \sum_{t=1}^T \varepsilon_t \tag{62}$$

when $\gamma_t \le \gamma$. A ratio form gives the familiar Cauchy–Schwarz variant: if $(\Delta_t^{\mathrm{AIR}}(p_t))^2 \le \Psi_t \mathcal{I}_t^{\mathrm{AIR}}(p_t)$, then

$$\mathbb{E}\,\mathrm{Reg}_T \le \sqrt{\left(\sum_{t=1}^T \mathbb{E}\Psi_t\right) H(A^\star)}.$$

Since $H(A^\star) \le \log n$, this route scales with the finite decision index and the sequential AIR coefficient. It therefore has a natural route to closing finite-action or low-dimensional action-set gaps that arise when GP–UCB's confidence multiplier is analyzed in the ambient RKHS. The route is different from GP–E2D: AMS/EBO uses the action-index posterior trajectory, whereas GP–E2D optimizes a one-round model-index offset whose relationship to the realized log determinant must be justified separately.

Position relative to the GP–UCB optimality debate. The original GP–UCB analysis is practical and trajectory-sensitive because it pays the realized information gain (53), or a worst-case relaxation only at the final step (Srinivas et al., 2010). Later work asks whether the remaining suboptimality comes from the algorithm or from the confidence analysis; in particular, Vakili et al. (2021b) ask for sharper online RKHS confidence intervals and sharper finite-action kernel-bandit guarantees. The indexed view separates the issue into three mechanisms. GP–UCB has the strongest realized-information geometry but may have a conservative MAIR coefficient. AMS/EBO can keep the action-index AIR telescope $I(A^\star; H_T) \le \log n$ and therefore has a principled route to finite-decision complexity, provided the robust bracket can be controlled. GP–E2D directly optimizes a one-round DEC coefficient, but an unlocalized DEC can be vacuous and a localized DEC must be matched to the same posterior-reference trajectory. Thus these algorithms should not be compared by a single universal coefficient; they are three relaxations of the same indexed Bellman identity with different estimation and coefficient mechanisms.

Why global RKHS DEC can be vacuous. Consider a Gaussian RKHS bandit with noise variance σ^2, feature map $\phi(x) \in \mathcal{H}$ satisfying $\|\phi(x)\|_{\mathcal{H}} \le 1$, and suppose the action set contains points $\{x_j : j \ge 1\}$ with $\phi(x_j) = e_j$, where $\{e_j\}_{j\ge1}$ is orthonormal in $\mathcal{H}$. Let the reference model be $f_0 \equiv 0$, and let $\mathcal{F}_B = \{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \le B\}$. For any one-round decision distribution p, define

$$q_j := \mathbb{E}_{x\sim p}\langle e_j, \phi(x)\rangle_{\mathcal{H}}^2.$$

Since

$$\sum_{j\geq 1} q_j \leq \mathbb{E}_{x\sim p}\|\phi(x)\|_{\mathcal{H}}^2 \leq 1,$$

there are directions with arbitrarily small q_j. For the alternative $f_j = Be_j$, the optimal action is x_j, while

$$\mathbb{E}_{x\sim p}\Delta_{f_j}(x) = B - B\,\mathbb{E}_{x\sim p}\langle e_j, \phi(x)\rangle_{\mathcal{H}} \geq B(1-\sqrt{q_j}),$$

and the one-round KL information against f_0 is

$$\mathbb{E}_{x\sim p}D_{\mathrm{KL}}\big(N(f_j(x),\sigma^2)\,\big\|\,N(0,\sigma^2)\big) = \frac{B^2}{2\sigma^2}q_j.$$

Hence, for every finite offset coefficient γ, one may choose j with q_j so small that

$$\mathbb{E}_{x\sim p}\Delta_{f_j}(x) - \gamma\mathbb{E}_{x\sim p}D_{\mathrm{KL}}\big(N(f_j(x),\sigma^2)\,\big\|\,N(0,\sigma^2)\big) \geq cB$$

for a numerical constant $c > 0$. The constrained version has the same pathology: for any fixed information radius, a sufficiently rare direction has information inside the radius while retaining regret of order B. Thus an unlocalized one-round DEC over the entire RKHS ball need not reflect the effective difficulty of the finite experiment being run. The meaningful object is the finite active marginal together with its posterior-reference trajectory. Indexed Bellman information complexity keeps this localization inside the state process, rather than appending a separate DEC localization after the fact.

GP–E2D offset must be localized. The model-index offset in (59) is meaningful only relative to a comparison set and a reference law. If the comparison set is the entire infinite-dimensional RKHS ball, then a one-round adversary can place mass in directions receiving arbitrarily small action probability. The resulting regret remains order one while the one-step information is arbitrarily small, making the global offset vacuous. If the comparison set is instead the finite active posterior neighborhood, the offset may be finite and useful. The point is that the localization is part of the representation and posterior-reference trajectory; it should not be appended as an independent estimation term after a separate log-determinant analysis.

5.2 MAB lower bound via optimal-action index and reference history

In this subsection, we apply our lower-bound approach to multi-armed bandits (MAB). Consider $K \geq 3$ Gaussian arms with unit noise variance. Let the ambient model space be a class $\Omega_{\mathrm{MAB}} \subseteq \mathbb{R}^K$ of mean vectors, and write a model environment as $\omega = (\theta_\omega(1), \ldots, \theta_\omega(K))$. Pulling arm a in environment ω produces an observation distributed as $N(\theta_\omega(a), 1)$. Let

$$A^\star(\omega) \in \arg\max_{a\in[K]} \theta_\omega(a)$$

with an arbitrary fixed tie-breaking rule, and define the information index

$$\chi(\omega) = A^\star(\omega) \in [K].$$

Thus the random environment is the full mean vector Ω, while the information target used in the lower certificate is only

$$Y = \chi(\Omega) = A^\star(\Omega).$$

For the lower bound, fix $\Delta > 0$ and use the K-point prior μ_Δ supported on the spike environments

$$\omega^j = \Delta e_j, \qquad j \in [K],$$

where e_j is the jth coordinate vector. Equivalently, draw $J \sim \mathrm{Unif}([K])$ and set $\Omega = \omega^J$. On this prior support the optimal arm is unique and

$$Y = \chi(\Omega) = J.$$

The equality $Y = J$ is only a property of this least-favorable prior; it should not be read as identifying the information target with the full model environment. The one-step regret in environment ω^j is

$$\ell_{\omega^j}(a) = \max_b \theta_{\omega^j}(b) - \theta_{\omega^j}(a) = \Delta \mathbb{1}\{a \neq j\}.$$

For brevity write $\ell_j = \ell_{\omega^j}$ on this subfamily.

Lemma 5.2 (MAB ghost entropy)**.** *Let $Y = \chi(\Omega)$ under the prior $\Omega \sim \mu_\Delta$. For every algorithm and every reference history H'_T independent of $Y \sim \mathrm{Unif}([K])$,*

$$\mathbb{P}\left(\bar{L}_Y(H'_T) \leq \frac{\Delta}{2}\right) \leq \frac{2}{K},$$

where $\bar{L}_y(H'_T) = T^{-1}\sum_{t=1}^{T} \ell_y(A'_t)$ and A'_t denotes the arm selected in the reference history at time t.

Proof. For a fixed reference action sequence $a'_1, \ldots, a'_T$, let

$$N'_y = \sum_{t=1}^{T} \mathbb{1}\{a'_t = y\}.$$

The average regret against the spike environment ω^y is

$$\bar{L}_y(H'_T) = \Delta(1 - N'_y/T).$$

The event $\bar{L}_y(H'_T) \leq \Delta/2$ implies $N'_y \geq T/2$. Since Y is uniform and independent of the reference sequence,

$$\mathbb{E}[N'_Y \mid H'_T] = \frac{1}{K}\sum_{y=1}^{K} N'_y = \frac{T}{K}.$$

Markov's inequality gives

$$\mathbb{P}(N'_Y \geq T/2 \mid H'_T) \leq \frac{2}{K}.$$

Averaging over H'_T proves the claim. $\square$

Lemma 5.3 (MAB information bound)**.** *There is a universal constant $c_0 > 0$ such that if $\Delta^2 T/K \leq c_0$, then every adaptive algorithm satisfies*

$$I(Y; H_T) \leq \frac{1}{4}\log(K/2),$$

where $Y = \chi(\Omega)$ and $\Omega \sim \mu_\Delta$.

Proof. Let $\mathbb{P}_0$ be the law of the history under the zero-mean reference model, let $\mathbb{P}_j$ be the law under the spike environment ω^j, and let

$$\bar{\mathbb{P}} = \frac{1}{K}\sum_{j=1}^{K}\mathbb{P}_j$$

be the marginal law of H_T under the prior μ_Δ. Under this prior, $Y = j$ is equivalent to $\Omega = \omega^j$, but the mutual information is still the action-index information $I(Y; H_T)$, not information about an ambient full mean vector outside the prior support. Since $\bar{\mathbb{P}} \geq K^{-1}\mathbb{P}_j$, the likelihood ratio $d\mathbb{P}_j/d\bar{\mathbb{P}}$ is bounded by K. The inequality

$$D_{\mathrm{KL}}(P\|Q) \leq (2 + \log K)H^2(P, Q) \qquad \text{whenever } Q \geq K^{-1}P,$$

together with the triangle inequality and convexity for squared Hellinger distance, gives

$$I(Y; H_T) = \frac{1}{K}\sum_{j=1}^{K} D_{\mathrm{KL}}(\mathbb{P}_j\|\bar{\mathbb{P}}) \leq 4(2 + \log K)\frac{1}{K}\sum_{j=1}^{K} H^2(\mathbb{P}_j, \mathbb{P}_0).$$

Using $H^2(P, Q) \leq D_{\mathrm{KL}}(Q\|P)$ and the adaptive Gaussian KL chain rule under the zero model,

$$\frac{1}{K}\sum_{j=1}^{K} H^2(\mathbb{P}_j, \mathbb{P}_0) \leq \frac{1}{K}\sum_{j=1}^{K} D_{\mathrm{KL}}(\mathbb{P}_0\|\mathbb{P}_j) = \frac{\Delta^2}{2K}\mathbb{E}_0\sum_{t=1}^{T}\sum_{j=1}^{K}\mathbb{1}\{A_t = j\} = \frac{\Delta^2 T}{2K}.$$

Therefore

$$I(Y; H_T) \leq 2(2 + \log K)\frac{\Delta^2 T}{K}.$$

Choosing, for example,

$$c_0 \leq \inf_{K\geq 3}\frac{\log(K/2)}{8(2 + \log K)}$$

which is a strictly positive universal constant, proves the displayed bound. □

Theorem 5.4 (MAB lower bound)**.** *For Gaussian K-armed bandits with unit noise variance, over any ambient mean-vector class Ω_{MAB} that contains the spike instances $\{\Delta e_j : j \in [K]\}$ used below,*

$$\mathfrak{R}_T^\star \geq c\sqrt{KT}$$

for a universal constant $c > 0$ whenever $T \geq K \geq 3$. In particular, this applies to any bounded class such as $[0, 1]^K$ once the universal constant in the choice of Δ is small enough.

Proof. Choose

$$\Delta = c_1\sqrt{K/T}$$

with c_1 small enough that Lemma 5.3 applies. Let $r = \Delta/2$. Lemma 5.2 gives $p_r \leq 2/K$, while Lemma 5.3 gives

$$T\bar{C} = I(Y; H_T) \leq \frac{1}{4}\log(K/2).$$

For c_1 small enough, condition (41) holds. Theorem 4.1, applied uniformly over algorithms, gives a Bayes regret lower bound under the prior μ_Δ supported on the spike subfamily. Since this prior is supported inside the ambient model class Ω_{MAB}, the minimax regret over Ω_{MAB} is at least this Bayes risk. Hence

$$\mathfrak{R}_T^\star \geq \frac{Tr}{2} = \frac{T\Delta}{4} = c\sqrt{KT}$$

for a universal constant $c > 0$. □

On the upper-bound side, the same distinction between the full model environment and the action index is important. An action-index AIR analysis uses $Y = \chi(\Omega) = A^\star(\Omega)$ as the information target, rather than the full environment Ω. This avoids charging information for distinctions between environments that do not change the optimal action. A model-index E2D relaxation can pay a K-factor both through model estimation and through the DEC term, whereas the action-index AIR identity in Theorem 3.5 incurs only a $\log K$ information-gain term and pays the K-factor through the AIR coefficient (Xu and Zeevi, 2025). This nearly matches the lower bound.

5.3 Linear bandits lower bound via optimal-action index and reference history

Consider stochastic linear bandits with action set

$$\mathcal{A} = \{a \in \mathbb{R}^d : \|a\|_2 \le 1\}.$$

Let the ambient model class be the Euclidean unit parameter ball

$$\Omega_{\text{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \le 1\},$$

and regard an environment $\omega \in \Omega_{\text{lin}}$ as the full mean parameter vector. At time t, after choosing $A_t \in \mathcal{A}$, the learner observes

$$O_t = \langle \omega, A_t \rangle + \xi_t, \qquad \xi_t \sim N(0, 1),$$

where the noises are independent over time and independent of the learner's external randomization. For $\omega \ne 0$, the unique optimal action over $\mathcal{A}$ is

$$A^\star(\omega) = \frac{\omega}{\|\omega\|_2}.$$

At $\omega = 0$, fix any deterministic tie-breaking action; this case is not used by the lower-bound prior. The information index is the optimal action

$$\chi(\omega) = A^\star(\omega), \qquad Y = \chi(\Omega),$$

not the full environment Ω. Thus two environments that have the same optimal action are not distinguished by the index unless the prior support itself makes them distinguishable.

We prove the standard $\Omega(d\sqrt{T})$ minimax lower bound through the quantile indexed information theorem. The proof uses a hypercube prior together with an adaptive information bound. The role of the hypercube variable below is only to code the finitely many optimal actions in the prior support. It should not be read as replacing the ambient environment Ω by a model label. Note that for a non-Gaussian prior, one cannot condition on the realized adaptive design matrix and then apply the fixed-design Gaussian-channel capacity formula, because the design itself is a statistic of the past observations and may carry information about the unknown environment and hence about Y. Chen et al. (2024, Section 3.3) prove a closely related lower bound by a different route: they use the interactive Fano method with a Gaussian prior and a unit-sphere calculation. Both approaches show that action-indexed information arguments can recover the sharp linear-bandit order, whereas model-indexed relaxations such as DEC can lose factors when they charge for estimating more of the model than is needed to identify the optimal action.

Let

$$V \sim \text{Unif}(\{\pm 1\}^d), \qquad \Omega = \Delta V.$$

Assume $\Delta\sqrt{d} \le 1$, so that this prior is supported on Ω_{lin}. On this prior support,

$$Y = \chi(\Omega) = A^\star(\Omega) = \frac{V}{\sqrt{d}}. \tag{63}$$

The map $v \mapsto v/\sqrt{d}$ is one-to-one on $\{\pm 1\}^d$. Hence, for this prior only, V may be used as a code for the action index Y, and

$$I(Y; H_T) = I(V; H_T).$$

For an index value $y = v/\sqrt{d}$, write ℓ_v for the regret under the corresponding environment $\omega = \Delta v$. The one-step regret of action $a \in \mathbb{R}^d$, $\|a\|_2 \le 1$, is

$$\ell_v(a) = \Delta\sqrt{d} - \Delta\langle v, a\rangle.$$

Assume $T \ge d^2$ and choose $\Delta \le 1/\sqrt{d}$, so that $\|\Omega\|_2 = \Delta\sqrt{d} \le 1$.

Lemma 5.5 (Linear ghost entropy)**.** *There is a universal constant $c_{\text{gh}} > 0$ such that every deterministic reference action sequence $a'_1, \ldots, a'_T$, with $\|a'_t\|_2 \le 1$, satisfies*

$$\mathbb{P}_{V\sim\text{Unif}(\{\pm1\}^d)}\left(\frac{1}{T}\sum_{t=1}^{T}\left(\Delta\sqrt{d} - \Delta\langle V, a'_t\rangle\right) \le \frac{\Delta\sqrt{d}}{4}\right) \le \exp(-c_{\text{gh}} d).$$

Consequently, the same bound holds after averaging over any random reference history H'_T whose induced action sequence is independent of the action index Y, equivalently independent of V under the hypercube prior.

Proof. Let

$$\bar{a} := \frac{1}{T}\sum_{t=1}^{T} a'_t.$$

Since $\|\bar{a}\|_2 \le T^{-1}\sum_t \|a'_t\|_2 \le 1$, the event in the display implies

$$\Delta\sqrt{d} - \Delta\langle V, \bar{a}\rangle \le \frac{\Delta\sqrt{d}}{4},$$

or equivalently,

$$\langle V, \bar{a}\rangle \ge \frac{3\sqrt{d}}{4}.$$

The random variable $\langle V, \bar{a}\rangle = \sum_{i=1}^d V_i \bar{a}_i$ is a centered Rademacher sum with variance proxy $\|\bar{a}\|_2^2 \le 1$. Hoeffding's inequality therefore gives

$$\mathbb{P}\left(\langle V, \bar{a}\rangle \ge \frac{3\sqrt{d}}{4}\right) \le \exp\left(-\frac{9d}{32}\right).$$

Thus the first claim holds, for example with $c_{\text{gh}} = 9/32$. If the reference action sequence is random but independent of Y, then because $Y = V/\sqrt{d}$ is a bijective function of V on the prior support, the reference sequence is independent of V. Conditioning on the reference history and applying the deterministic bound gives the second claim. □

Lemma 5.6 (Adaptive hypercube information capacity)**.** *For the hypercube prior above, every possibly randomized adaptive algorithm satisfies*

$$I(Y;H_T) = I(V;H_T) \le \Delta^2 T,$$

where $H_T = (A_1, O_1, \ldots, A_T, O_T)$.

Proof. Since $Y = V/\sqrt{d}$ is a bijective function of V on the hypercube support, $I(Y;H_T) = I(V;H_T)$. It remains to bound $I(V;H_T)$. Let U denote the learner's internal random seed, independent of V. Then

$$I(V;H_T) \le I(V;H_T,U) = \mathbb{E}_U I(V;H_T \mid U).$$

It is therefore enough to prove the claim after conditioning on U, so that the algorithm is deterministic.

For $v \in \{\pm1\}^d$, let $\mathbb{P}_v$ denote the trajectory law when $\Omega = \Delta v$. We use the chain rule over coordinates:

$$I(V;H_T) = \sum_{i=1}^{d} I(V_i;H_T \mid V_1, \ldots, V_{i-1}).$$

Fix a coordinate i and a prefix $u = (v_1, \ldots, v_{i-1})$. Let $\mathbb{P}_{u,+}$ and $\mathbb{P}_{u,-}$ be the trajectory laws after averaging uniformly over the remaining coordinates $V_{i+1}, \ldots, V_d$, conditional respectively on $V_i = +1$ and $V_i = -1$. Then

$$I(V_i;H_T \mid V_{<i} = u) = D_{\mathrm{JS}}(\mathbb{P}_{u,+}, \mathbb{P}_{u,-}),$$

where D_{JS} denotes Jensen–Shannon divergence with equal weights. For any two probability measures P, Q,

$$D_{\mathrm{JS}}(P,Q) \le \frac{1}{4}\Big(D_{\mathrm{KL}}(P\|Q) + D_{\mathrm{KL}}(Q\|P)\Big).$$

Next couple the two mixtures by using the same suffix. If $w \in \{\pm1\}^{d-i}$ and

$$v^+(u,w) = (u, +1, w), \qquad v^-(u,w) = (u, -1, w),$$

then joint convexity of relative entropy gives

$$D_{\mathrm{KL}}(\mathbb{P}_{u,+}\|\mathbb{P}_{u,-}) \le \mathbb{E}_w D_{\mathrm{KL}}\big(\mathbb{P}_{v^+(u,w)} \,\big\|\, \mathbb{P}_{v^-(u,w)}\big),$$

and the same argument gives the analogous reverse inequality.

For two fixed vertices v^+ and v^- that differ only in coordinate i, the adaptive KL chain rule gives

$$\begin{aligned} D_{\mathrm{KL}}(\mathbb{P}_{v^+}\|\mathbb{P}_{v^-}) &= \frac{1}{2}\mathbb{E}_{v^+} \sum_{t=1}^{T} \Big(\Delta\langle v^+ - v^-, A_t\rangle\Big)^2 \\ &= 2\Delta^2 \mathbb{E}_{v^+} \sum_{t=1}^{T} A_{t,i}^2. \end{aligned}$$

Here the action kernel contributes no KL because, after conditioning on the learner's internal randomness, the algorithm uses the same decision rule under both environments; only the Gaussian observation kernel changes. Similarly,

$$D_{\mathrm{KL}}(\mathbb{P}_{v^-}\|\mathbb{P}_{v^+}) = 2\Delta^2 \mathbb{E}_{v^-} \sum_{t=1}^{T} A_{t,i}^2.$$

Combining the preceding displays yields

$$
\begin{aligned}
I(V_i; H_T \mid V_{<i} = u) &\le \frac{\Delta^2}{2}\mathbb{E}_w\left[\mathbb{E}_{v^+(u,w)}\sum_{t=1}^T A_{t,i}^2 + \mathbb{E}_{v^-(u,w)}\sum_{t=1}^T A_{t,i}^2\right] \\
&= \Delta^2 \mathbb{E}\left[\sum_{t=1}^T A_{t,i}^2 \,\middle|\, V_{<i} = u\right],
\end{aligned}
$$

where the final expectation is under the original hypercube prior and the algorithm's trajectory law. Averaging over $V_{<i}$ and summing over i gives

$$
\begin{aligned}
I(V; H_T) &\le \Delta^2 \mathbb{E}\sum_{t=1}^T \sum_{i=1}^d A_{t,i}^2 \\
&= \Delta^2 \mathbb{E}\sum_{t=1}^T \|A_t\|_2^2 \\
&\le \Delta^2 T.
\end{aligned}
$$

This proves the claim. □

Remark 5.7 (Why the fixed-design log-determinant argument is not used)**.** *For a Gaussian prior on a full parameter vector, posterior conjugacy gives the adaptive information identity*

$$
I(\Theta; H_T) = \frac{1}{2}\mathbb{E}\log\det\left(I + \Sigma_0^{1/2}\Big(\sum_{t=1}^T A_t A_t^\top\Big)\Sigma_0^{1/2}\right),
$$

and this is bounded by a trace-constrained log determinant. For the hypercube prior, however, the posterior is not Gaussian and the realized design matrix is itself a statistic of the observations. Conditioning only on that realized design matrix controls the conditional observation information given the design; it does not by itself control the information carried by the adaptive design. Lemma 5.6 therefore uses a coordinatewise adaptive KL argument to control $I(Y; H_T)$ directly.

Theorem 5.8 (Linear bandit lower bound)**.** *For stochastic linear bandits with action set $\{a \in \mathbb{R}^d : \|a\|_2 \le 1\}$, unit Gaussian noise, and unit-ball parameter class $\Omega_{\text{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \le 1\}$,*

$$
\mathfrak{R}_T^\star \ge c\, d\sqrt{T}
$$

for a universal constant $c > 0$, whenever $T \ge d^2$.

Proof. Assume first that d is larger than a sufficiently large universal constant d_0 to be fixed below. Choose

$$
\Delta = c_1\sqrt{\frac{d}{T}},
$$

where $c_1 > 0$ is a sufficiently small universal constant. Since $T \ge d^2$, choosing $c_1 \le 1$ ensures

$$
\Delta\sqrt{d} = c_1\frac{d}{\sqrt{T}} \le c_1 \le 1,
$$

so the hypercube prior is supported on the unit parameter ball.

Let

$$r = \frac{\Delta\sqrt{d}}{4}.$$

Lemma 5.5 gives the reference ghost-good probability bound

$$p_r \le \exp(-c_{\mathrm{gh}} d).$$

Lemma 5.6 gives the adaptive action-index information bound

$$T\bar{C} = I(Y; H_T) = I(V; H_T) \le \Delta^2 T = c_1^2 d.$$

Choose c_1 so small that $c_1^2 \le c_{\mathrm{gh}}/4$, and then choose d_0 so large that $\log 2 \le c_{\mathrm{gh}} d/4$ for all $d \ge d_0$. Then, for all $d \ge d_0$,

$$T\bar{C} + \log 2 \le \frac{1}{2} \log \frac{1}{p_r}.$$

Theorem 4.1, applied to the action index $Y = \chi(\Omega) = A^\star(\Omega)$, gives a Bayes regret lower bound under the hypercube prior of at least

$$\frac{Tr}{2} = \frac{T\Delta\sqrt{d}}{8} = \frac{c_1}{8} d\sqrt{T}.$$

Since this prior is supported on Ω_{lin}, the minimax regret over Ω_{lin} is at least this Bayes risk.

It remains to handle the finitely many dimensions $1 \le d < d_0$. For these dimensions, use the two-point prior $\Omega = \sigma\delta e_1$, where $\sigma \sim \mathrm{Unif}(\{\pm 1\})$, e_1 is the first coordinate vector, and $\delta = (2\sqrt{T})^{-1}$. This prior is also supported on the unit parameter ball. Under $\sigma = +1$, the optimal action is e_1; under $\sigma = -1$, the optimal action is $-e_1$. Let $\mathbb{P}_+$ and $\mathbb{P}_-$ be the laws of the history under these two environments, and let $\mathbb{P}_\pm^{t-1}$ denote the law of the history before action t. For any algorithm,

$$\begin{aligned}
\frac{1}{2}\mathbb{E}_+\left[\delta(1 - A_{t,1})\right] + \frac{1}{2}\mathbb{E}_-\left[\delta(1 + A_{t,1})\right] &= \delta\left(1 - \frac{1}{2}(\mathbb{E}_+ A_{t,1} - \mathbb{E}_- A_{t,1})\right) \\
&\ge \delta\left(1 - \mathrm{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1})\right).
\end{aligned}$$

By Pinsker's inequality and the adaptive Gaussian KL chain rule,

$$\mathrm{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1}) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(\mathbb{P}_+^{t-1} \| \mathbb{P}_-^{t-1})} \le \delta\sqrt{t-1} \le \frac{1}{2}.$$

Therefore each round has Bayes regret at least $\delta/2$, and the total Bayes regret is at least $T\delta/2 = \sqrt{T}/4$. Since $d < d_0$, this is at least $(4d_0)^{-1} d\sqrt{T}$. Reducing the universal constant c, if necessary, completes the proof for all d. □

5.4 Bellman-rank as indexed representation

Low Bellman rank was introduced by Jiang et al. (2017) as a sufficient structural condition under which contextual decision processes with suitable bilinear Bellman-error structure are PAC-learnable. In the present language, a Bellman-rank factorization can help build a low-dimensional Bellman-sufficient representation when it is supplemented by predictive, update, loss, and index sufficiency; the factorization alone controls Bellman-error evaluation and does not, by itself, determine the full observation process. The following proposition is structural; it is not a new sharp lower bound for every Bellman-rank class.

Proposition 5.9 (Bellman rank with predictive sufficiency gives an indexed representation)**.** *Suppose a contextual decision process admits an exact Bellman-error factorization of rank d in the sense of Jiang et al. (2017): for the stages and hypothesis–policy pairs in the planning class, the relevant Bellman error can be written as an inner product of two d-dimensional witness-feature vectors. Fix a prior μ and an index map χ. If the observation model is dominated by a common σ-finite measure and if, conditional on the current posterior or calibrated confidence object, the predictive law, the posterior-reference update, the relevant loss or benchmark terms, and the conditional χ-indexed predictive laws and losses in Definition 2.3 are all determined by these Bellman-error features together with the retained planning variables, then the process admits an indexed Bellman sufficient representation with state that may be chosen as a posterior or calibrated confidence object over the d-dimensional witness coordinates, augmented by the planning variables needed for Bellman updates. Consequently, whenever one constructs a valid information supersolution and a valid ghost-good-mass supersolution for the induced feature experiment, for example in a linear-Gaussian or uniformly bounded-likelihood setting, Theorem 4.3 yields a Bellman quantile-index lower certificate.*

Proof. The Bellman-error factorization supplies d-dimensional coordinates sufficient to evaluate the one-step Bellman-error terms covered by the factorization. It does not by itself imply predictive sufficiency, loss sufficiency, update sufficiency, or index sufficiency. These are precisely the additional assumptions imposed in the proposition. If the predictive observation law and posterior update are functions of this statistic and the current posterior, and if the relevant losses, benchmarks, and χ-conditional predictive objects are functions of the same retained state, then Definition 2.3 is satisfied with S_t equal to the posterior or calibrated confidence object over these features together with the accumulated planning state. The information and ghost-good Bellman recursions are then well-defined on the same state. Applying Theorem 4.3 to any valid pair of supersolutions for this state space gives the claimed certificate. The proposition is intentionally conditional because Bellman-rank learnability is primarily a structural upper-bound statement; sharp lower bounds require a packing or ghost-entropy calculation inside the particular rank-d class. $\square$

This application clarifies the connection between the present framework and the broader perspective on representation learning developed in Li and Xu (2026); Xu (2026). The point is not that every low-Bellman-rank problem has the same minimax rate or that Bellman rank alone determines a lower bound. The point is that a Bellman-rank factorization can provide a candidate state on which the upper supersolution and the lower quantile-information certificate should be compared, after the necessary predictive, update, loss, and index-sufficiency conditions have been verified.

6 Conclusion

The paper argues that the natural unit of information complexity in interactive decision making is a Bellman representation equipped with an indexed posterior-reference process. Once the state and index are chosen, the same objects appear on both sides of the theory. Upper bounds are regret supersolutions or potential identities whose brackets are paid for by indexed information. Lower bounds are quantile-information certificates: the reference trajectory supplies both the information telescope and the ghost probability of small average regret. The critical radius is determined by

$$T\bar{C}_\chi \asymp \log \frac{1}{p_r},$$

which is the interactive analogue of the classical local-prior-mass/Fano balance.

AIR and MAIR are two choices of index within the same log-gain Bellman supersolution. Choosing $\chi(\Omega) = A^\star(\Omega)$ gives action-index AIR, which is often the right description of decision difficulty. Choosing $\chi(\Omega) = \Omega$, or the active reward vector in a bandit problem, gives model-index MAIR, which is often the convenient description of estimation geometry. UCB, E2D/DEC, and AMS/EBO then become three relaxations of the same indexed bracket: calibration and optimism, one-step offset optimization, and robust saddle control.

The kernel-bandit application illustrates why this separation matters. GP–UCB is practical because it keeps the realized log-determinant information gain of the actual trajectory. AMS/EBO, in an AIR form, can scale with the entropy of the finite or low-dimensional optimal-action index and the sequential AIR bracket. GP–E2D optimizes a one-round DEC offset, but its interaction with realized information gain is a theorem to be proved rather than a formal consequence of the offset definition; without localization a global RKHS DEC may be vacuous. Thus the right comparison is not a universal conversion coefficient. It is whether a single Bellman-sufficient state supports matching upper supersolutions and lower quantile certificates without charging estimation difficulty twice.

Several directions remain. The first is to identify representations for which the regret supersolution and the Bellman-Fano lower certificate match up to constants or logarithms. The second is to replace one-step DEC calculations by dynamically consistent Bellman relaxations when posterior evolution matters. The third is to sharpen the finite-action kernel theory by proving which of GP–UCB, AMS/EBO, or GP–E2D attains the finite-marginal information scale under the weakest calibration assumptions. These are representation-level questions. The goal is not a universal one-shot notion of dimension, but a distribution-dependent dynamic theory of state spaces that accounts for estimation, decision, and lower-bound entropy costs once, rather than repeatedly.

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